



OpenShift Cluster Manager 1-latest

Managing clusters

Using Red Hat OpenShift Cluster Manager to work with your OpenShift clusters

OpenShift Cluster Manager 1-latest Managing clusters

Using Red Hat OpenShift Cluster Manager to work with your OpenShift clusters

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Abstract

You can use Red Hat OpenShift Cluster Manager to work with your Red Hat OpenShift cloud services and Red Hat OpenShift Container Platform clusters. With OpenShift Cluster Manager you can create, subscribe, and manage different types of OpenShift clusters from a single user interface.

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CHAPTER 1. WHAT IS RED HAT OPENSIFT CLUSTER MANAGER?

Red Hat OpenShift Cluster Manager is a managed service on the [Red Hat Hybrid Cloud Console](#) where you can create, operate, and upgrade your Red Hat OpenShift 4 clusters.

OpenShift Cluster Manager provides links and steps to install Red Hat OpenShift Container Platform clusters and tools to create Red Hat OpenShift Dedicated and Red Hat OpenShift Service on AWS (ROSA) clusters.

From OpenShift Cluster Manager, you can work with all of your organization's OpenShift Container Platform and OpenShift cloud services clusters from a single dashboard. You can gain insights and recommendations for managing your clusters, and you can complete the following tasks:

- View high level cluster information.
- Create new clusters.
- Configure Red Hat subscription services on your clusters.
- Manage your clusters using other services on the [Red Hat Hybrid Cloud Console](#).
- Monitor clusters for problems.
- Access the OpenShift cluster admin console
- Find information about the latest OpenShift versions.
- Download tools to use with your clusters.
- Get support for your clusters and manage your Red Hat support cases.

Additional resources

- [OpenShift Container Platform documentation](#)
- [OpenShift Dedicated documentation](#)
- [ROSA documentation](#)
- [Subscriptions documentation](#)
- [{red-hat-insights} remote health monitoring with connected clusters](#)

1.1. GETTING STARTED WITH OPENSIFT CLUSTER MANAGER

You can use Red Hat OpenShift Cluster Manager to work with your Red Hat OpenShift cloud services and Red Hat OpenShift Container Platform clusters. With OpenShift Cluster Manager you can create, subscribe, and manage different types of OpenShift clusters from a single user interface.

Prerequisites

- You have a Red Hat account.

Procedure

- Enter the following URL in a web browser:

<https://console.redhat.com/openshift/>



NOTE

For details about web browser requirements, see the **Browser Support** link at the bottom of the [Red Hat Hybrid Cloud Console](#) landing page.

1.2. WHAT IS THE DIFFERENCE BETWEEN OPENSIFT CONTAINER PLATFORM AND OPENSIFT DEDICATED?

Red Hat OpenShift Container Platform is a self-managed hybrid cloud platform that you can deploy on-premises, on bare metal, or on public cloud infrastructure.

OpenShift Dedicated is a fully managed service running on AWS or Google Cloud Platform (GCP), where Red Hat handles the cluster operations on your behalf.

Regardless of your deployment model, you can use OpenShift Cluster Manager to create and manage both your OpenShift Container Platform and OpenShift Dedicated clusters from a single, unified dashboard.

Additional resources

- [Red Hat OpenShift product page](#)
- [OpenShift deployment methods](#)

1.3. USING OPENSIFT CLUSTER MANAGER WITH OPENSIFT CONTAINER PLATFORM

You can use the OpenShift Cluster Manager user interface to create OpenShift Container Platform clusters and subscribe the clusters to Red Hat for support. OpenShift Cluster Manager provides the installer and instructions to create self-managed clusters on each supported environment for OpenShift Container Platform. You can then view and manage your OpenShift Container Platform clusters in OpenShift Cluster Manager, or log in to the OpenShift Container Platform web console to access and configure your clusters.



NOTE

You cannot log in to your Red Hat account with your email unless your email is your Red Hat login. For more information, see [Finding your login](#).

You can find information about the latest OpenShift Container Platform release versions available, as well as update channels for your clusters from the **Releases** menu in OpenShift Cluster Manager. For insights about your clusters, use the integrated services within the [Red Hat Hybrid Cloud Console](#) such as Red Hat Lightspeed Advisor, Subscriptions, and Cost Management.

Additional resources

- [OpenShift Container Platform documentation](#)

1.4. USING OPENSIFT CLUSTER MANAGER WITH OPENSIFT DEDICATED

You can use the OpenShift Cluster Manager user interface to create, view, and manage your OpenShift Dedicated clusters.

OpenShift Dedicated clusters are managed by Red Hat and are known as *managed clusters*. You can create OpenShift Dedicated clusters on AWS or Google Cloud Platform, using either the Red Hat managed cloud account or your own cloud provider account. When you use your own cloud provider account, this billing model is referred to as *Customer Cloud Subscription (CCS)* in OpenShift Cluster Manager.

Additional resources

- [OpenShift Dedicated documentation](#)

1.5. USING OPENSIFT CLUSTER MANAGER WITH RED HAT OPENSIFT SERVICE ON AWS

You can use the OpenShift Cluster Manager user interface to create, view and manage your Red Hat OpenShift Service on AWS (ROSA) clusters. ROSA is a fully-managed OpenShift service, jointly managed and supported by Red Hat and Amazon Web Services (AWS). This service is procured directly from your AWS aROSAnt. ROSA pricing is consumption based and is billed directly to your AWS account.

You can quickly deploy ROSA from OpenShift Cluster Manager or using the ROSA CLI. In OpenShift Cluster Manager, you can manage your ROSA cluster and any add-on services.

Additional resources

- [ROSA documentation](#)

1.6. USING OPENSIFT CLUSTER MANAGER AND THE RED HAT HYBRID CLOUD CONSOLE

OpenShift Cluster Manager is integrated with the following services hosted on the [Red Hat Hybrid Cloud Console](#), which you can use to gain deeper understanding and manage your OpenShift Container Platform clusters.

- [Red Hat Lightspeed Advisor](#) for OpenShift Container Platform monitors the health of your OpenShift Container Platform clusters and helps you identify, prioritize, and resolve risks to service availability, fault tolerance, performance, and security.
- [Subscriptions](#) is a service that you can use to monitor your usage and subscription information for your OpenShift Container Platform clusters.
- [Cost management](#) aggregates and displays the costs of your OpenShift Container Platform deployment and infrastructure across bare-metal servers, virtual machines, private clouds, and public cloud infrastructure, including Amazon Web Services and Microsoft Azure.

You need a Red Hat account to access OpenShift Cluster Manager and the [Red Hat Hybrid Cloud Console](#). You can then deploy an OpenShift Container Platform cluster in OpenShift Cluster Manager.

**NOTE**

You cannot log in to your Red Hat account with your email unless your email is your Red Hat login. For more information, see [Finding your login](#).

For greater security, you can use two-factor authentication (2FA) to access OpenShift Cluster Manager and the [Red Hat Hybrid Cloud Console](#). You must enable 2FA in your Red Hat account to use 2FA to access OpenShift Cluster Manager. Organization Administrators can enable 2FA for all users in their organization, or individual users can configure 2FA for their own Red Hat account.

To enable 2FA in your Red Hat account or learn more, see the [Using Two-Factor Authentication](#) guide.

Additional resources

- [Remote health monitoring with connected clusters](#)
- [Subscriptions documentation](#)
- [Cost management documentation](#)
- [Red Hat Hybrid Cloud Console documentation](#)
- [Create a Red Hat Login](#)

1.7. ADD-ON SERVICES WITH YOUR OPENSIFT CLOUD SERVICES CLUSTERS

Add-ons are additional services that you can install to your existing Red Hat OpenShift Dedicated and Red Hat OpenShift Service on AWS (ROSA) clusters to enhance cluster capabilities.

You can install and manage add-on services from a cluster's **Add-ons** tab in [OpenShift Cluster Manager](#).

Depending on the add-on service, you might need additional Red Hat subscriptions or quota to use it. See the documentation for the add-on to learn more about the requirements and for instructions for using the add-on.

- To learn about add-ons for OpenShift Dedicated, see [Add-on services available for OpenShift Dedicated](#).
- To learn about add-ons for ROSA, see [Add-on services available for Red Hat OpenShift Service on AWS](#).

Additional resources

- [Managing your add-on services](#)

1.8. OPENSIFT CLUSTER NOTIFICATIONS

Configure notification settings to monitor OpenShift cluster events effectively. You can manage default email notifications and integrate third-party services, such as Slack, Google Chat, and Microsoft Teams, for cluster event alerts. By default, you will receive email notifications about OpenShift cluster events.

**NOTE**

All options are not available for all services. You cannot disable OpenShift notifications on this page because your cluster is managed by Red Hat. These notifications are the primary way that Red Hat Site Reliability Engineering (SRE) will contact you to inform you about cluster problems and request actions you must take to resolve them. Cluster owners cannot unsubscribe from email notifications. If you are not a cluster owner and you do not want to receive notification emails, you can ask your cluster owner or administrator to remove you from the list of cluster notification contacts as described in [Removing notification contacts from your cluster](#).

In addition to email notifications, Red Hat Hybrid Cloud Console Organization Administrators, Cloud Administrators, or users with Notifications Administrator permissions can configure cluster event notifications through third-party products such as Slack, Google Chat, and Microsoft Teams.

For more information about OpenShift Cluster Manager notifications integrations, see [Integrating the Red Hat Hybrid Cloud Console with third-party applications](#).

CHAPTER 2. MANAGING THE CLUSTER LIFECYCLE

You can use Red Hat OpenShift Cluster Manager to create and delete OpenShift clusters, and manage the cluster lifecycle.

You can create different types of OpenShift clusters from [OpenShift Cluster Manager](#).

2.1. CREATING AN OPENSIFT CONTAINER PLATFORM CLUSTER

OpenShift Container Platform clusters run on your own infrastructure. Using OpenShift Container Platform, you can create your clusters on a private or public cloud, or on bare metal, using the command-line installer.

Create your cluster using OpenShift Cluster Manager and the installation program for your environment or cloud account. After provisioning your cluster, configure your Red Hat subscription to get support for your cluster.

Prerequisites

- A Red Hat login.
- Your own on-premises infrastructure. For example, a platform such as Red Hat Virtualization or Red Hat OpenStack; a cloud provider such as AWS; or a bare-metal Linux machine.

Procedure

1. Go to [OpenShift Cluster Manager](#) and click **Create cluster** on the Red Hat OpenShift Container Platform tile.
2. Follow the instructions provided in the OpenShift Cluster Manager user interface to create your OpenShift cluster.
 - + By default, your cluster automatically registers to the OpenShift Cluster Manager service the first time your cluster boots after installation and is connected to the Telemetry service. The cluster registers with a 60-day evaluation subscription that does not include Red Hat support.
 - +

You can also create an on-premises cluster using the Assisted Installer. For more information about this method, see [Installing on-premise with Assisted Installer](#).

Verification

- After you provision your cluster, click **Cluster List** list in the left navigation panel in OpenShift Cluster Manager and confirm that your cluster is listed.

Next steps

After creating your OpenShift Container Platform cluster, you can use the link Red Hat Hybrid Cloud Console to:

- Configure your Red Hat subscription to get support for your cluster from the **Edit subscription settings** menu in OpenShift Cluster Manager. You can subscribe your OpenShift Container Platform clusters to an Annual Red Hat subscription or an On-Demand subscription from the Red Hat Marketplace.

Additional resources

Additional resources

- [OpenShift Container Platform installation documentation](#)
- [Remote health monitoring with connected clusters](#)
- [Configuring OpenShift Container Platform cluster subscriptions](#)
- [Red Hat Lightspeed Advisor recommendations](#)
- [Subscribing an OpenShift Container Platform cluster](#)

2.2. CREATING AN OPENSIFT DEDICATED CLUSTER

OpenShift Dedicated clusters are managed by Red Hat and provisioned on Amazon Web Services (AWS) or Google Cloud Platform (GCP). They are referred to as *managed clusters*.

Using OpenShift Cluster Manager, you can create an OpenShift Dedicated cluster on AWS or GCP using a cloud account owned by Red Hat or with your own cloud account using the *Customer Cloud Subscription* (CCS) model.

When creating your OpenShift Dedicated cluster, you must also configure the Red Hat subscription type for the cluster to use. Your cluster is then automatically subscribed to your Red Hat subscriptions and includes Premium-level support.

You can use the following types of Red Hat subscriptions to support your OpenShift Dedicated clusters:

- **Annual:** A subscription providing a fixed capacity of resources, pre-purchased from Red Hat. Cluster provisioning is based on available quota. Quota is allocated from your Red Hat subscriptions and is required to scale up a cluster. See [Section 2.2.1, "Creating an OpenShift Dedicated cluster with an annual subscription"](#).
- **On-Demand:** A subscription allowing flexible usage, billed through the Red Hat Marketplace. When you enable an On-Demand subscription in Red Hat Marketplace, you set resource limits for your services to control usage automatically. See [Section 2.2.2, "Creating an OpenShift Dedicated cluster with an On-Demand subscription"](#).
- **OpenShift Dedicated trial:** You can try OpenShift Dedicated for 60 days free of charge with a trial Red Hat subscription. You can upgrade your cluster to a paid Red Hat subscription at any time. See [About the OpenShift Dedicated Trial](#) and [Section 2.2.3, "Creating an OpenShift Dedicated cluster with a free trial Red Hat subscription"](#) for more details.

You can view your quota and resource limits alongside cluster usage, based on your active OpenShift Dedicated clusters, from the [Subscriptions](#) area in OpenShift Cluster Manager.



IMPORTANT

You cannot change the subscription type on an existing OpenShift Dedicated cluster, with the exception of upgrading a trial subscription. For detailed instructions about creating a new cluster, see [Creating a cluster](#) in the OpenShift Dedicated documentation.

Additional resources

- [Getting started](#)
- [Managing OpenShift Dedicated cluster subscriptions](#)

- [Red Hat Marketplace](#)
- [About the OpenShift Dedicated Trial](#)

2.2.1. Creating an OpenShift Dedicated cluster with an annual subscription

You can create an OpenShift Dedicated cluster using an **annual** (fixed capacity) Red Hat subscription on your own AWS or Google Cloud Platform cloud provider (Customer Cloud Subscription), or use the Red Hat fully-managed cloud to run your OpenShift Dedicated clusters.

This is the traditional Red Hat subscription type and support is pre-purchased from Red Hat and billed annually. Cluster provisioning is based on available quota. Quota is allocated from Red Hat subscriptions and is required to scale up a cluster.

To use an On-Demand (flexible usage) subscription for your cluster instead, see [Section 2.2.2, “Creating an OpenShift Dedicated cluster with an On-Demand subscription”](#).



IMPORTANT

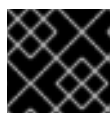
You cannot change the subscription type after the cluster is created.

Prerequisites

- A Red Hat login
- Your organization must have an active Red Hat OpenShift Dedicated subscription with sufficient quota to create a cluster. Check your available quota from [Subscriptions > Dedicated \(Annual\)](#).
- If you are creating a cluster on your own AWS or Google Cloud (Customer Cloud Subscription), you must configure your cloud account before creating your cluster. See the OpenShift Dedicated [Planning your environment](#) guide to understand the requirements for Customer Cloud Subscriptions.

Procedure

1. From [OpenShift Cluster Manager](#), click **Create cluster**.
2. From the **Cloud** tab next to **OpenShift Dedicated**, click **Create cluster**. You can also view your available quota from this screen before creating your cluster.
3. In the **Billing Model** screen, select **Annual** as your **Subscription type**.



IMPORTANT

You cannot change the subscription type after the cluster is created.

4. Select your **Infrastructure type**:
 - **Customer Cloud Subscription** uses your own cloud account, where you control billing and Red Hat manages the cluster for you. You must configure your cloud account before creating your cluster. See the OpenShift Dedicated [Planning](#) guide for instructions.
 - **Red Hat cloud account** deploys your cluster in cloud provider accounts owned by Red Hat. For this option, Red Hat handles all billing and management for your cluster.

5. Configure basic cluster settings, including your cloud provider, machine pools, networking, and update policies.

**NOTE**

See the [OpenShift Dedicated documentation](#) for detailed information about configuring your cluster settings.

6. Click **Create cluster** to provision your cluster.

Verification

- While your cluster is being provisioned, you can view it in the **Clusters** list in OpenShift Cluster Manager. The cluster shows its **Status** as **Ready** when provisioning is complete.

Next steps

After creating your OpenShift Dedicated cluster and its status is **Ready**, you can:

- Configure an identity provider to set up user access. For instructions, see [Configuring identity providers](#).
- Configure privileged **dedicated-admin** and **cluster-admin** users to access your cluster. Learn more about OpenShift Dedicated roles in [Managing administration roles and users](#).
- Upgrade your cluster. To learn more, see [OpenShift Dedicated cluster upgrades](#).
- View logs and notifications about your cluster. For more information, see [Logging](#).
- Create machine pools in your cluster. For more information, see [About machine pools](#).
- Use the monitoring stack to check your projects in isolation from Red Hat Site Reliability Engineer (SRE) platform metrics. For more information, see [Understanding the monitoring stack](#).
- Monitor the health of your cluster. For more information, see [About remote health monitoring](#).
- Scale your OpenShift Dedicated cluster up or down, if you have purchased the necessary subscriptions to do so. For instructions, see [Scaling your cluster](#).
- Install add-ons to expand the capabilities of your cluster. For more information, see [Adding a service to a cluster](#).
- View a summary of the resource limits and usage for your cluster in the Red Hat Hybrid Cloud Console at [Subscriptions > Dedicated \(Annual\)](#).

Additional resources

- [Getting started](#)
- [Managing OpenShift Dedicated cluster subscriptions](#)

2.2.2. Creating an OpenShift Dedicated cluster with an On-Demand subscription

You can use the **On-Demand** subscription type to create an OpenShift Dedicated cluster that is billed by usage. Billing is post-paid and handled by the Red Hat Marketplace.

When you enable an On-Demand subscription in Red Hat Marketplace, you set resource limits for your services to control maximum usage automatically.

This option allows the flexibility to scale your cluster up and down as needed, and to be billed accordingly. You can track usage by cluster in [Subscriptions > Dedicated \(On-Demand\)](#).

To use an annual (fixed capacity) subscription for your cluster instead, see [Section 2.2.1, “Creating an OpenShift Dedicated cluster with an annual subscription”](#).



IMPORTANT

You cannot change the subscription type after the cluster is created.

Prerequisites

- A Red Hat login
- Your organization must have an active OpenShift Dedicated On-Demand subscription enabled from the Red Hat Marketplace with resource limits configured.
 - Check your resource limits from the [Subscriptions > Dedicated \(On-Demand Limits\)](#) area in OpenShift Cluster Manager. If no resource limits display, go to [Red Hat Marketplace](#) to enable your OpenShift Dedicated On-Demand subscription.
- Your own AWS or Google Cloud account, configured as described in the OpenShift Dedicated [Planning your environment](#) documentation.

Procedure

1. From [OpenShift Cluster Manager](#), click **Create cluster**.
2. From the **Cloud** tab next to **OpenShift Dedicated**, click **Create cluster**. You can also view your available quota from this screen before creating your cluster.
3. In the **Billing Model** screen, select **On-Demand** as your **Subscription type**.



NOTE

If **On-Demand** is not available as an option, follow the prompts in the OpenShift Cluster Manager user interface to enable an account and link your billing information in the [Red Hat Marketplace](#). You can also verify whether On-Demand billing (Red Hat Marketplace subscriptions) is configured from the [Subscriptions > Dedicated \(On-Demand Limits\)](#) in OpenShift Cluster Manager.

You cannot change the subscription type after the cluster is created.

4. Select **Customer Cloud Subscription** as your **Infrastructure type**. This is required for the On-Demand subscription option.



NOTE

The **Customer Cloud Subscription** option uses your own cloud account, where you control billing and Red Hat manages the cluster for you. You must configure your cloud account before creating your cluster. See the OpenShift Dedicated [Planning](#) guide for instructions.

5. Configure basic cluster settings, including your cloud provider, cluster details, machine pools, networking, update policies.

**NOTE**

See the [OpenShift Dedicated documentation](#) for detailed information about configuring your cluster settings.

6. Click **Create cluster** to provision your cluster.

Verification

- While your cluster is being provisioned, you can view it in the **Clusters** list in OpenShift Cluster Manager. The cluster shows its **Status** as **Ready** when provisioning is complete.

Next steps

After creating your OpenShift Dedicated cluster and its status is **Ready**, you can:

- Configure an identity provider to set up user access. For instructions, see [Configuring identity providers](#).
- Configure privileged **dedicated-admin** and **cluster-admin** users to access your cluster. Learn more about OpenShift Dedicated roles in [Managing administration roles and users](#).
- Upgrade your cluster. To learn more, see [OpenShift Dedicated cluster upgrades](#).
- View logs and notifications about your cluster. For more information, see [Logging](#).
- Create machine pools in your cluster. For more information, see [About machine pools](#).
- Use the monitoring stack to check your projects in isolation from Red Hat Site Reliability Engineer (SRE) platform metrics. For more information, see [Understanding the monitoring stack](#).
- Monitor the health of your cluster. For more information, see [About remote health monitoring](#).
- Scale your OpenShift Dedicated cluster up or down, if you have purchased the necessary subscriptions to do so. For instructions, see [Scaling your cluster](#).
- Install add-ons to expand the capabilities of your cluster. For more information, see [Adding a service to a cluster](#).
- View a summary of your cluster usage in the Red Hat Hybrid Cloud Console at [Subscriptions > Dedicated \(On-Demand\)](#).
- View a summary of the resource limits for your cluster in the Red Hat Hybrid Cloud Console at [Subscriptions > Dedicated \(On-Demand Limits\)](#).

Additional resources

- [Getting started](#)
- [Managing OpenShift Dedicated cluster subscriptions](#)
- [Red Hat Marketplace documentation](#)

- [Subscriptions documentation](#)

2.2.3. Creating an OpenShift Dedicated cluster with a free trial Red Hat subscription

You can try OpenShift Dedicated free of charge for 60 days by using a trial Red Hat subscription.

OpenShift Dedicated trial clusters are provided without an uptime service level agreement (SLA) and are self-supported.

To use a free trial subscription for your cluster, you need to use your own AWS or Google Cloud account (Customer Cloud Subscription) to provide the infrastructure.

At any time before the 60-day trial is finished, you can upgrade your cluster with your own Red Hat subscription details to continue using OpenShift Dedicated. At the end of the 60-day trial period, if you have not upgraded the trial cluster, your OpenShift Dedicated trial cluster and all installed add-on services are marked for permanent deletion.

To find out more and sign up for an OpenShift Dedicated trial subscription, go to [About the OpenShift Dedicated Trial](#) or [Try OpenShift](#).

Prerequisites

- A Red Hat login
- An OpenShift Dedicated trial subscription enabled. See [About the OpenShift Dedicated Trial](#) for details.
- Your own AWS or Google Cloud account, configured as described in the OpenShift Dedicated [Planning your environment](#) documentation.

Procedure

1. From [OpenShift Cluster Manager](#), click **Create cluster**.
2. From the **Cloud** tab next to **Red Hat OpenShift Dedicated Trial**, click **Create trial cluster**. You can also view your available quota from this screen before creating your cluster.
 - In the next screen (**Billing model**), **Free trial (upgradeable)** will be automatically selected as your **Subscription type**, and **Customer Cloud Subscription** will be automatically selected as your **Infrastructure type**.



NOTE

- If **Free trial (upgradeable)** is not available as an option, follow the prompts in the user interface to enable your Red Hat account.
- **Customer Cloud Subscription** uses your own cloud account, where you control billing and Red Hat manages the cluster for you. You must configure your cloud account before creating your cluster. See the OpenShift Dedicated [Planning](#) guide for instructions.

3. Configure basic cluster settings, including your cloud provider, machine pools, networking, and update policies.

**NOTE**

See the [OpenShift Dedicated documentation](#) for detailed information about configuring your cluster settings.

4. Click **Create cluster** to provision your cluster.

Verification

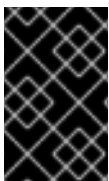
- While your cluster is being provisioned, you can view it in the **Clusters** list in OpenShift Cluster Manager. The cluster will show its **Status** as **Ready** when provisioning is complete.

See the cluster overview page to view the expiry date of the trial subscription and other details. This information also shows on the **Clusters** list in the **Created** column.

Next steps

After creating your OpenShift Dedicated cluster and its status is **Ready**, you can:

- Configure an identity provider to set up user access. For instructions, see [Configuring identity providers](#).
- Configure privileged **dedicated-admin** and **cluster-admin** users to access your cluster. Learn more about OpenShift Dedicated roles in [Managing administration roles and users](#).
- Upgrade your cluster. To learn more, see [OpenShift Dedicated cluster upgrades](#).
- View logs and notifications about your cluster. For more information, see [Logging](#).
- Create machine pools in your cluster. For more information, see [About machine pools](#).
- Use the monitoring stack to check your projects in isolation from Red Hat Site Reliability Engineer (SRE) platform metrics. For more information, see [Understanding the monitoring stack](#).
- Monitor the health of your cluster. For more information, see [About remote health monitoring](#).
- Scale your OpenShift Dedicated cluster up or down, if you have purchased the necessary subscriptions to do so. For instructions, see [Scaling your cluster](#).
- Install add-ons to expand the capabilities of your cluster. For more information, see [Adding a service to a cluster](#).
- View a summary of the resource limits and usage for your cluster in the Red Hat Hybrid Cloud Console at [Subscriptions > Dedicated \(Annual\)](#).

**IMPORTANT**

Before your 60-day trial expires, upgrade your cluster to a paid fully-supported Red Hat subscription to continue using your cluster. See [Upgrading an OpenShift Dedicated trial cluster to a fully supported cluster](#).

Additional resources

- [Try OpenShift](#)

- [About the OpenShift Dedicated Trial](#)
- [Getting started](#)
- [Managing OpenShift Dedicated cluster subscriptions](#)
- [Red Hat OpenShift Dedicated](#)

2.3. CREATING A RED HAT OPENSIFT SERVICE ON AWS

Red Hat OpenShift Service on AWS (ROSA) is a fully-managed OpenShift service, jointly managed and supported by both Red Hat and Amazon Web Services (AWS). Having your clusters maintained by this service gives you the freedom to focus on deploying applications.

This service is licensed directly from your AWS account. ROSA pricing is consumption-based and is billed directly to your AWS account.

You can create a ROSA cluster from OpenShift Cluster Manager or by using the ROSA CLI.

To create a ROSA cluster in OpenShift Cluster Manager, go to [the cluster creation page](#), find **Red Hat OpenShift Service on AWS** and click **Create cluster**.

For full details about creating a ROSA cluster from OpenShift Cluster Manager or the ROSA CLI, see [Getting Started](#) in the ROSA documentation.

Additional resources

- [Product documentation for Red Hat OpenShift Service on AWS](#)

2.4. UPDATING OPENSIFT CLOUD SERVICES CLUSTERS

OpenShift Cluster Manager allows you to update, or upgrade, your cloud services clusters. Updating your cluster to the latest version ensures your cluster is secure and supported. Cluster updates are rolling per node and do not require you to take your cluster or workloads offline.

The **Overview** and **Settings** tabs for your cluster in OpenShift Cluster Manager contain notifications if an OpenShift version update is available for your Red Hat OpenShift Dedicated (OSD) or Red Hat OpenShift Service on AWS (ROSA) cluster. From the **Settings** tab, you can apply updates to your cluster, or configure your update strategy for future cluster updates.

You can choose to schedule recurring or individual updates for your cluster by selecting one of these update strategy options in the **Settings** tab:

- **Individual updates** allows you to schedule one update at a time manually.
- **Recurring updates** allows you to configure a schedule for updates to occur at a preferred time every week if an update is available. Recurring updates only occur when a new patch (z-stream) update becomes available at least two days prior to your selected start time. Recurring updates only schedule minor (y-stream) updates if the update has been manually acknowledged in the **Settings** tab of OpenShift Cluster Manager.

You can configure either update strategy for both minor version and patch (z-stream) updates.

**NOTE**

In case of Critical security concerns (CVEs) that significantly impact the security or stability of the cluster, updates might be automatically scheduled by Red Hat Site Reliability Engineering (SRE) to the latest z-stream version not impacted by the CVE within two [business days](#) after customer notifications if the customer has not already scheduled an update.

Additional resources

- [Upgrading OpenShift Dedicated](#)
- [Upgrading Red Hat OpenShift Service on AWS](#)
- [Updating clusters](#)

2.5. REMOVING AN OPENSIFT CONTAINER PLATFORM

You can archive an OpenShift Container Platform cluster to remove it from [OpenShift Cluster Manager](#). Archiving a cluster removes it from subscription management and from the cluster list in OpenShift Cluster Manager.

You cannot delete an OpenShift Container Platform cluster from your infrastructure using OpenShift Cluster Manager.

**NOTE**

To fully delete an OpenShift Container Platform cluster, see the instructions for destroying a cluster on your infrastructure type in the OpenShift Container Platform [Installing](#) documentation.

Prerequisites

- A Red Hat login..
- An OpenShift Container Platform cluster..
- You must have the Cluster Owner or Cluster Editor role on the cluster, or Organization Administrator privileges in your Red Hat account..

Procedure

1. In [OpenShift Cluster Manager](#), select the cluster that that you want to archive.
2. Click  (**more options**) > **Archive cluster** to open the archiving dialog.
3. Click **Archive cluster** to confirm.

Verification

When archiving is complete, your cluster no longer appears in the **Clusters** list in OpenShift Cluster Manager.

You can view all archived clusters in the [Cluster Archives](#) list in OpenShift Cluster Manager.



NOTE

You can restore an OpenShift Cluster Manager cluster from the archive by locating it in <https://console.redhat.com/openshift/archived> and clicking **Unarchive** next to the cluster. It appears in the **Clusters** list after it is unarchived.

Additional resources

- [Installing](#)

2.6. DELETING AN OPENSIFT DEDICATED CLUSTER FROM OPENSIFT CLUSTER MANAGER

You can delete OpenShift Dedicated clusters using [OpenShift Cluster Manager](#).

Prerequisites

- A Red Hat login.
- An OpenShift Dedicated cluster.
- You must have the Cluster Owner or Cluster Editor role on the cluster, or Organization Administrator privileges in your Red Hat account.

Procedure

1. In [OpenShift Cluster Manager](#), select the cluster that you want to delete.
2. Click  (more options) > **Delete cluster** to open the **Delete cluster** dialog.



WARNING

This action cannot be undone. It uninstalls the cluster, and all data is deleted.

3. Confirm that you want to delete the cluster by entering the cluster name in the dialog field and then click **Delete**.

Verification

- Your cluster shows **Uninstalling** in the **Status** column on the **Clusters** page.
- While the cluster deletion is in progress, you can view **Uninstallation logs** by opening the cluster details **Overview** page.

When the deletion is complete, your cluster no longer appears in the **Clusters** list in OpenShift Cluster Manager.

You can view your deleted clusters from the **Clusters** list by clicking  (more options) > **View cluster archives**.

2.7. DELETING A RED HAT OPENSIFT SERVICE ON AWS

You can delete Red Hat OpenShift Service on AWS (ROSA) clusters using [OpenShift Cluster Manager](#).

Prerequisites

- A Red Hat login.
- A Red Hat OpenShift Service on AWS (ROSA) cluster.
- You must have the Cluster Owner or Cluster Editor role on the cluster, or Organization Administrator privileges in your Red Hat account.

Procedure

1. In [OpenShift Cluster Manager](#), select the cluster that you want to delete.
2. Click  (more options) > **Delete cluster** to open the **Delete cluster** dialog.



WARNING

This action cannot be undone. It uninstalls the cluster, and all data is deleted.

3. Confirm that you want to delete the cluster by entering the cluster name in the dialog field and then click **Delete**.

Verification

- Your cluster shows **Uninstalling** in the **Status** column on the **Clusters** page.
- While the cluster deletion is in progress, you can view **Uninstallation logs** by opening the cluster details **Overview** page.

When the deletion is complete, your cluster no longer appears in the **Clusters** list in OpenShift Cluster Manager.

You can view your deleted clusters from the **Clusters** list by clicking  (more options) > **View cluster archives**.

2.8. VIEWING ARCHIVED AND DELETED CLUSTERS

You can view all deleted and archived OpenShift clusters in your organization from the [Cluster Archives](#) list in OpenShift Cluster Manager.

You can fully delete OpenShift Dedicated clusters in OpenShift Cluster Manager, but you can only archive OpenShift Container Platform clusters. Archiving an OpenShift Container Platform cluster removes it from the OpenShift Cluster Manager cluster list and from subscription management.

Prerequisites

- A Red Hat login

Procedure

- Click [Cluster Archives](#) to view a list of deleted and archived clusters in OpenShift Cluster Manager.



NOTE

You can restore an OpenShift Container Platform cluster from the archive by finding the cluster in <https://console.redhat.com/openshift/archived> and clicking **Unarchive** next to the cluster. It appears in the **Clusters** list after it is unarchived.

Additional resources

- [Installing and configuring OpenShift Container Platform clusters](#)

CHAPTER 3. CLUSTER SUBSCRIPTIONS AND REGISTRATION

Register your clusters to Red Hat OpenShift Cluster Manager and subscribe to Red Hat subscription management to access all Red Hat OpenShift functionality.

After you create a cluster, it is automatically registered to OpenShift Cluster Manager and subscribed to a Red Hat subscription where you can access Red Hat support and updates.

- OpenShift Container Platform clusters are automatically subscribed to a limited 60-day Red Hat evaluation subscription when created. To upgrade your OpenShift Container Platform cluster to your own Red Hat subscription, see [Section 3.1, “Configuring OpenShift Container Platform cluster subscriptions”](#).
- To create an OpenShift Dedicated cluster, you must enter your Red Hat subscription details at creation. You can use your own Red Hat subscription when creating an OpenShift Dedicated cluster, or try OpenShift Dedicated using a 60-day trial subscription to be upgraded later. To configure the subscription settings on your OpenShift Dedicated clusters, see [Section 3.2, “Managing OpenShift Dedicated cluster subscriptions”](#).
- For Red Hat OpenShift Service on AWS (ROSA) clusters, no subscription configuration is necessary in OpenShift Cluster Manager, because this service is licensed directly from your AWS account. ROSA pricing is consumption-based and is billed directly to your AWS account.

Additional resources

- [Subscriptions > Container Platform](#)
- [Subscriptions documentation](#)
- [Section 3.3, “Registering OpenShift Container Platform clusters to OpenShift Cluster Manager”](#)
- [Section 3.3.1, “Verifying your OpenShift Container Platform”](#)

3.1. CONFIGURING OPENSIFT CONTAINER PLATFORM CLUSTER SUBSCRIPTIONS

By default, your OpenShift Container Platform cluster automatically registers to the OpenShift Cluster Manager service (<https://console.redhat.com/openshift/>) after it is created, and is subscribed to a limited 60-day Red Hat evaluation subscription for access to Red Hat support and updates.

To avoid downtime on your cluster, you must edit the cluster subscription settings in OpenShift Cluster Manager to use your own Red Hat subscription *before* your evaluation subscription expires.

You can use one of the following Red Hat subscription types to support your OpenShift Container Platform clusters:

- **Annual:** A subscription that provides a fixed capacity of resources, pre-purchased from Red Hat.
- **On-Demand:** This subscription that allows flexible usage and is billed through the Red Hat Marketplace. Your clusters must be connected to Telemetry in OpenShift Cluster Manager to use this subscription type.
When you enable an On-Demand subscription in Red Hat Marketplace, you set resource limits for your services to control usage automatically.

You can view your active OpenShift Container Platform subscriptions from the [Subscriptions](#) area in OpenShift Cluster Manager.



NOTE

If you disabled Telemetry or your cluster cannot connect to api.openshift.com, you can alternatively complete the Red Hat registration process online at <https://console.redhat.com/openshift/register>. See [Registering disconnected clusters](#) for more information.

Prerequisites

- A Red Hat login
- An OpenShift Container Platform cluster
- A Red Hat Annual subscription or an OpenShift Container Platform On-Demand subscription enabled from the Red Hat Marketplace with resource limits configured.
 - Check your resource limits from the [Subscriptions](#) area in OpenShift Cluster Manager. If no resource limits display, enable On-Demand subscriptions in Red Hat Marketplace.
- You must have the Cluster Owner or Cluster Editor role on the cluster, or Organization Administrator privileges in your Red Hat account to edit a cluster's subscription settings.

Procedure

1. View an OpenShift Container Platform cluster in [OpenShift Cluster Manager](#).
2. Click **Actions** > **Edit subscription settings**. You can also access this from the **Subscription settings** section of the cluster **Overview** page.
3. Select your **Subscription type**:
 - **Annual** to use your Red Hat fixed capacity subscription
 - **On-Demand** to use your Red Hat Marketplace flexible usage subscription



IMPORTANT

If **On-Demand** is not available as an option, contact [Red Hat Sales](#) to enable an account and link your billing information in the Red Hat Marketplace. You can also verify whether On-Demand billing is enabled if **Marketplace** shows as **Enabled** in the [Subscriptions](#) area under **OpenShift Container Platform**.

After setting your subscription type, you cannot change a cluster's subscription type from On-Demand to an Annual subscription.

4. If you selected **Annual**, select the options that apply to your Red Hat subscription in the dialog. See [OpenShift Container Platform cluster subscription settings](#) for more details about the available settings. If your cluster is disconnected, the subscription type is automatically set to **Annual**. For **On-Demand**, all other settings are pre-configured.
5. Click **Save settings**.

It might take up to two hours for these settings to update for your cluster in the **Subscriptions** summary and Red Hat Subscription Management after making changes to your Red Hat subscription in the Red Hat Customer Portal.

Verification steps

View your cluster's subscription status and usage in [Subscriptions > Container Platform](#).

Additional resources

- For more details about Subscriptions, see the [Subscriptions documentation](#).
- Learn more about remote health monitoring with Telemetry and the Insights Operator in the [OpenShift Container Platform documentation](#).
- For more information about On-Demand subscriptions and billing, see the [Red Hat Marketplace documentation](#).

3.1.1. Changing OpenShift Container Platform

To ensure that you receive the correct level of support for your OpenShift Container Platform clusters, configure your cluster subscription settings in OpenShift Cluster Manager to align with the values for your Red Hat subscription.

For OpenShift Dedicated clusters, no further subscription configuration is needed after choosing your subscription type. OpenShift Dedicated clusters are automatically subscribed to Red Hat subscriptions and come with Premium-level support.



NOTE

To find more details about your subscriptions, view your [Subscriptions Inventory](#) in the Red Hat Customer Portal.

The following options are available for OpenShift Container Platform clusters:

Table 3.1. Subscription settings

Subscription setting	Options	Summary	More information
Subscription type	• Annual: Fixed capacity subscription from Red Hat • On-Demand: Flexible usage billed through the Red Hat Marketplace	What type of subscription are you using for this cluster?	Contact Red Hat Sales to enable On-Demand subscriptions from the Red Hat Marketplace for OpenShift Container Platform clusters.

Subscription setting	Options	Summary	More information
Service level agreement (SLA)	- Premium - Standard - Self-Support 60-day evaluation	How is this cluster supported?	The hours of coverage, support ticket response times, and other terms that are defined by the Service Level Agreement (SLA). See Production Support Terms of Service .
Support type	- Red Hat support (L1-L3) - Partner support (L3)	Which team do you contact for primary support?	If you purchased the subscription through Red Hat, select L1-L3.
Cluster usage	- Production - Development/Test - Disaster Recovery	How do you intend to use this cluster?	Describe the purpose of the cluster - for example, running production workloads or testing internal development projects.
Subscription units	- Cores/vCPUs - Sockets	How is usage measured for your subscription?	Define how the cluster will consume the purchased Red Hat subscriptions.
Number of compute cores	A value between 1-999	How many compute cores does your cluster use?	NOTE: Configuration is only necessary on disconnected clusters. This defines the number of physical cores used by your cluster. This excludes control plane nodes and nodes that do not use RHCOS. See Section 3.1.3, "Calculating the number of compute cores for your cluster" for more details.

Procedure

1. To change the support type of an OpenShift Container Platform cluster after it has been

initialized on Red Hat OpenShift Cluster Manager, click  (more options) > **Edit subscription settings** for a cluster.

3.1.2. Subscription usage for OpenShift Container Platform

The [subscriptions service](#) in the Red Hat Hybrid Cloud Console tracks Red Hat OpenShift usage as *cluster size* on physical and virtual systems. The cluster size is the sum of all subscribed nodes. A *subscribed node* is a compute or worker node that runs workloads, as opposed to a control plane or infrastructure node that manages the cluster.

The subscriptions service tracks Red Hat OpenShift Container Platform usage in physical CPU cores or sockets for clusters and aggregates this data into an account view.

Subscriptions are used based on a 1 physical core or 2 vCPU core rate.

As an example, the subscription *Red Hat OpenShift Container Platform, Standard/Premium (64 Cores or 128 vCPUs)* supports an environment using either of these measurements:

- 64 physical cores
- 32 physical cores and 64 vCPUs

Both configurations are counted the same. This information can be viewed in the **Cores** field of the subscriptions service usage record on the [Red Hat Hybrid Cloud Console](#).

Additional resources

- [Self-managed Red Hat OpenShift sizing and subscription guide](#)
- [Red Hat OpenShift](#)
- [Measurement of usage and capacity for Red Hat OpenShift](#)

3.1.3. Calculating the number of compute cores for your cluster

OpenShift Container Platform subscriptions are measured by physical cores and vCPUs or sockets, depending on your subscription terms.



NOTE

It is only necessary to configure the number of compute cores in OpenShift Cluster Manager for disconnected clusters.

In most cases, when configuring the **Number of compute cores** on your OpenShift Container Platform clusters, enter the number of compute cores and vCPUs or sockets included in your Red Hat subscription, excluding any control plane nodes. This applies to all clusters using Red Hat Enterprise Linux CoreOS (RHCOS) as an operating system.

However, if your clusters do not have any worker nodes, or the nodes are using Red Hat Enterprise Linux only (rather than RHCOS), usage for those nodes is not calculated against your OpenShift Container Platform subscription. In this case, enter a value of **1** for **Number of compute cores** for these nodes.

To confirm the unit of measurement used by your subscription and other details, log in to the [Red Hat Subscription Management](#) area in the Red Hat Customer Portal.



NOTE

You cannot log in to your Red Hat account with your email unless your email is your Red Hat login. For more information, see [Finding your login](#).

3.1.4. Obtaining CPU information for your cluster

Identify the CPU information required to calculate worker node subscriptions for your OpenShift Container Platform cluster. Use the **oc** command line tool to retrieve the necessary data.

Prerequisites

- An OpenShift Container Platform cluster
- Cluster administrator permissions
- OpenShift Container Platform **oc** CLI tool installed

Procedure

1. Log in to your OpenShift Container Platform cluster, enter the following command, and replace <username>, <password>, and <your-openshift-server> with the credentials for your OpenShift server:

```
$ oc login -u=<username> -p=<password> --server=<your-openshift-server> --insecure-skip-tls-verify
```

2. Obtain the roles of the nodes within the cluster (master, infrastructure or worker):

```
$ oc get nodes -o wide
```

3. Obtain the CPU information which is used to calculate the necessary subscriptions to cover worker nodes:

```
$ oc describe nodes | egrep 'Name:|InternalIP:|cpu:'
```

4. In [OpenShift Cluster Manager](#), enter the returned value in the **Number of compute cores** field for your cluster:
 - a. In OpenShift Cluster Manager, locate your cluster in the **Clusters** list.
 - b. Click  (more options) > **Edit subscription settings** for the cluster.
 - c. Enter the value in the **Number of compute cores** field and save the changes. You can now use your OpenShift usage data in the Red Hat Hybrid Cloud Console [subscriptions](#) service to inform your subscription renewal or purchase. Viewing your OpenShift hosts and nodes in OpenShift Cluster Manager, Red Hat Subscription Management, and the inventory in the Red Hat Hybrid Cloud Console shows the number of cores or vCPUs as known by the host.



NOTE

If you are running OpenShift on a hypervisor and your compute core count appears to be incorrect, ensure that you configure your hypervisor so that it groups vCPUs as your subscription is arranged. See this [Knowledgebase article](#) for more information.

Additional resources

- [Red Hat OpenShift](#)
- [Measurement of usage and capacity for Red Hat OpenShift](#)
- [Using Notifications and Reports with Red Hat Subscription Management](#)
- [Using Red Hat Subscription Management](#)
- [Red Hat Enterprise Linux CoreOS \(RHCOS\) documentation](#)

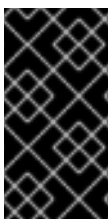
3.2. MANAGING OPENSIFT DEDICATED CLUSTER SUBSCRIPTIONS

OpenShift Cluster Manager allows you to create OpenShift Dedicated clusters using your Red Hat subscriptions. These clusters are managed by Red Hat and come with Premium-level support. You must enter your subscription details when creating an OpenShift Dedicated cluster.

You can use one of the following types of Red Hat subscriptions to create OpenShift Dedicated clusters:

- **Annual:** A subscription providing a fixed capacity of resources, pre-purchased from Red Hat. Cluster provisioning is based on available quota. Quota is allocated from your Red Hat subscriptions and is required to scale up a cluster.
- **On-Demand:** A subscription allowing flexible usage, billed through the Red Hat Marketplace. When you enable an On-Demand subscription in [Red Hat Marketplace](#), you set resource limits for your services to control usage automatically.
- **OpenShift Dedicated trial:** You can try OpenShift Dedicated for 60 days free of charge with a trial Red Hat subscription. You can upgrade your cluster to a paid Red Hat subscription at any time. See [About the OpenShift Dedicated Trial](#) for more details.

You can view your quota and resource limits, based on your active OpenShift Dedicated clusters, from the [Subscriptions](#) menu in OpenShift Cluster Manager.



IMPORTANT

You must select the subscription type when creating the cluster. You cannot change the subscription type on an existing OpenShift Dedicated cluster, with the exception of upgrading a trial subscription. To create a new cluster, see [Creating an OpenShift Dedicated cluster](#).

Prerequisites

- A Red Hat login
- An active Red Hat OpenShift Dedicated subscription with sufficient quota to create a cluster. See <https://www.openshift.com/products/dedicated/> for more information.

- You must have the Cluster Owner or Cluster Editor role on the cluster, or Organization Administrator privileges in your Red Hat account to edit a cluster's subscription settings.

Procedure

- To view a summary of all subscriptions for OpenShift Dedicated purchased by your organization or granted by Red Hat, go to OpenShift Cluster Manager and click **Subscriptions**. The summary also shows how much of your quota and resource limits have been used by your OpenShift Dedicated clusters, broken down by subscription type.
 - For Red Hat annual subscriptions, you can check your usage and quota in [Subscriptions > Dedicated \(Annual\)](#). You can also find out more about your subscription details in the [Red Hat Customer Portal](#) and purchase more quota if desired.



NOTE

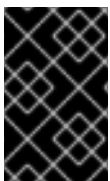
It might take up to two hours for your cluster's subscription status to update in OpenShift Cluster Manager after making changes in the Red Hat Customer Portal.

- For Red Hat Marketplace On-Demand subscriptions, you can check your total usage and usage by cluster in [Subscriptions > Dedicated \(On-Demand\)](#). You can view your resource limits in [Subscriptions > Dedicated \(On-Demand Limits\)](#). You can also view the total usage for your On-Demand OpenShift Dedicated clusters in Red Hat Marketplace. For information about billing for On-Demand subscriptions, see [Pay as you go products](#) in the Red Hat Marketplace documentation.
- For trial subscriptions, follow the prompts to upgrade your cluster to a paid Red Hat subscription before your trial finishes. See [About the OpenShift Dedicated Trial](#) and [Section 3.2.1, "Upgrading an OpenShift Dedicated trial cluster to a fully supported cluster"](#) for more details.

3.2.1. Upgrading an OpenShift Dedicated trial cluster to a fully supported cluster

You can upgrade your OpenShift Dedicated (OSD) trial cluster at any time after starting the free trial.

You can choose to upgrade your trial cluster before the trial conclusion if you want to run production services or use features that are not included in the free trial, such as autoscaling, specific add-on services, and quota increases.



IMPORTANT

The OpenShift Dedicated free trial ends when you delete your cluster or after 60 days, whichever happens first. At that time, your OpenShift Dedicated trial cluster and all installed add-on services are marked for permanent deletion.

If you upgrade the cluster before the trial is over, you can continue using the resources you created during the trial without interruption.

Prerequisites

- A Red Hat login
- An OpenShift Dedicated cluster using a trial subscription

- A Red Hat subscription for OpenShift Dedicated
- You must have the Cluster Owner or Cluster Editor role on the cluster, or Organization Administrator privileges in your Red Hat account to edit a cluster's subscription settings.

Procedure

1. Go to the **Clusters** list in [OpenShift Cluster Manager](#).
2. Find your OpenShift Dedicated trial cluster, labeled **OSD Trial** in the **Cluster Type** column.
3. Click **Upgrade from trial** and follow the instructions to upgrade your cluster.



NOTE

If the **Upgrade from trial** option does not appear, the reasons could include the following:

- You do not have the permissions needed to upgrade this cluster to a fully supported cluster. You must be an Organization Administrator on the Red Hat account or the Cluster Owner to upgrade the account.
- This cluster account is already upgraded to a fully supported OpenShift Dedicated cluster.

Verification steps

- Find your OpenShift Dedicated cluster in the **Clusters** list in [OpenShift Cluster Manager](#). The **Cluster type** is no longer be listed as **OSD Trial**.

Additional resources

- For more information about subscription types for OpenShift Dedicated clusters, see [Managing OpenShift Dedicated cluster subscriptions](#).
- For more details about OpenShift Dedicated trial clusters, see:
 - [About the OpenShift Dedicated Trial](#)
 - [Try OpenShift](#)

3.3. REGISTERING OPENSIFT CONTAINER PLATFORM CLUSTERS TO OPENSIFT CLUSTER MANAGER

To monitor the health of your OpenShift Container Platform clusters with [Insights Advisor](#) and receive alerts, updates, and recommendations from Red Hat Insights, your clusters must be registered to OpenShift Cluster Manager and subscribed to a Red Hat subscription.

By default, every OpenShift Container Platform cluster automatically registers to OpenShift Cluster Manager the first time the cluster boots after installation.

OpenShift Container Platform clusters report health and usage data to Red Hat through Telemetry and the Insights Operator when registered in OpenShift Cluster Manager. These are referred to as *connected clusters*.

Occasionally an OpenShift Container Platform cluster does not automatically register to the OpenShift Cluster Manager service (referred to as a *disconnected cluster*), for example if:

- the cluster was created in an air-gapped environment and cannot reach OpenShift Cluster Manager to inform OpenShift Cluster Manager it has been created
- you disabled the Telemeter client
- your cluster cannot connect to **api.openshift.com**

In this situation, you can register a disconnected cluster to OpenShift Cluster Manager manually from <https://console.redhat.com/openshift/register>. You can also enter your Red Hat subscription details from here to subscribe your cluster to Red Hat support.

After an OpenShift Container Platform cluster is registered and subscribed, you can then monitor your subscription capacity and usage in [Subscriptions > Container Platform](#).

3.3.1. Verifying your OpenShift Container Platform

You can verify that your OpenShift Container Platform cluster is registered to OpenShift Cluster Manager and subscribed to a Red Hat subscription from OpenShift Cluster Manager.

A OpenShift Container Platform cluster that is registered on OpenShift Cluster Manager is referred to as a *connected cluster*. In rare cases, for example, if Telemetry is disabled or blocked on the user's network, the cluster cannot be registered automatically and you must manually register the cluster to OpenShift Cluster Manager.

Prerequisites

- A Red Hat login
- An OpenShift Container Platform cluster
- To edit a cluster's subscription settings, you must have the Cluster Owner or Cluster Editor role on the cluster, or Organization Administrator privileges in your Red Hat account.

Procedure

1. Go to the **Clusters** list in [OpenShift Cluster Manager](#) and locate your OpenShift Container Platform cluster.



NOTE

If your cluster does not appear in the Clusters list, you need to register your cluster. See [Registering disconnected clusters](#) for instructions.

2. Review the **Status** column for the cluster:
 - If the **Status** is **Ready**, it is connected to OpenShift Cluster Manager and reporting Telemetry data. No manual registration is required.
 - If the **Status** is **Disconnected**, it is not sending Telemetry data to OpenShift Cluster Manager. This is due to the cluster being installed on a private network, or having Telemetry disabled.

- If the **Status** is **Stale**, your cluster is connected but has not sent Telemetry data to OpenShift Cluster Manager recently.
3. Review the **Created** column for the cluster to see the cluster subscription status:
- A date: Your cluster is subscribed to a Red Hat subscription and is receiving support and updates.
 - **60-day evaluation**: Your cluster is subscribed to Red Hat support with a temporary evaluation subscription. Configure the cluster to access Red Hat support with your own Red Hat subscription by clicking  (more options) > **Edit subscription settings**.
 - **Evaluation expired**: Your cluster is not subscribed to Red Hat support. Configure your subscription details for the cluster by clicking  (more options) > **Edit subscription settings**.



NOTE

- You can also check the cluster's subscription settings from the cluster **Overview** page.
- You must have the Cluster Owner or Cluster Editor role on the cluster, or Organization Administrator privileges in your Red Hat account to edit a cluster's subscription settings.

Next steps

- If your cluster is not registered (a *disconnected cluster*) to OpenShift Cluster Manager, register it with the steps in [Registering disconnected clusters](#).
- If your cluster is not subscribed to a Red Hat subscription or you need to update your subscription settings, see [Configuring OpenShift Container Platform cluster subscriptions](#).

3.3.1.1. Registering disconnected clusters

To monitor the health of your OpenShift Container Platform clusters and receive alerts, updates, and recommendations from Red Hat Lightspeed Advisor, your clusters must be registered to OpenShift Cluster Manager. If your cluster does not appear on the **Clusters** list in [OpenShift Cluster Manager](#), you need to register it.

If your cluster is already registered to OpenShift Cluster Manager and you only want to edit subscription settings for your cluster, click the **More options** icon, then **Edit subscription settings**, or configure your subscription settings from the cluster details page. See [Configuring OpenShift Container Platform cluster subscriptions](#) for details.

Prerequisites

- A Red Hat login
- An OpenShift Container Platform cluster
- A Red Hat subscription

- You must have the Cluster Owner or Cluster Editor role on the cluster, or Organization Administrator privileges in your Red Hat account, to edit cluster subscription settings.

Procedure

1. Go to the **Clusters** list in [OpenShift Cluster Manager](#).

To register a disconnected cluster, create a profile for your cluster manually in OpenShift Cluster Manager:

1. Go to the **Clusters** list in [OpenShift Cluster Manager](#).
2. At the top of the **Clusters** list, click the **More options** icon, then **Register cluster** to open the [Register disconnected cluster](#) page.
3. Enter the **Cluster ID** for the cluster you want to register. For example, `00000c9e-f75e-44e4-86e1-ebf60ec0b000`. You can find your cluster ID in the cluster web console in OpenShift Container Platform.
4. Enter the **Display name** for the cluster. This can be any name that you want to identify the cluster by in OpenShift Cluster Manager. You can find your cluster by this name in the cluster list for your organization.
5. Enter the **Web console URL** for the cluster. This is the URL to log into your OpenShift Container Platform cluster web console.
6. To subscribe your cluster to Red Hat support, enter your Red Hat subscription details in **Subscription settings**:
 - a. Select the **Service level agreement (SLA)** for the cluster.
 - b. Select your **Support type**.
 - c. Specify how you intend to use the cluster in **Cluster usage**.
 - d. Specify the unit your subscription is measured in (cores/vCPUs or sockets) in **Subscription units**.
 - e. Enter the number of compute cores your cluster will consume.
7. Click **Register cluster** to confirm registration and subscription.



NOTE

See [OpenShift Container Platform cluster subscription settings](#) for details about the subscription settings in OpenShift Cluster Manager.

Verification

1. Find your cluster displayed in the **Clusters** list in [OpenShift Cluster Manager](#).
2. The subscription configuration displays in the **Subscription settings** section. This can now be edited.
3. Go to [Subscriptions > Container Platform](#) to verify you can view subscription information about your clusters, including capacity and subscription usage.

Additional resources

- [{red-hat-insights} remote health monitoring with connected clusters](#)
- [{red-hat-insights} remote health monitoring with connected clusters](#)
- [OpenShift Container Platform documentation](#)
- [OpenShift Container Platform documentation](#)
- [Subscriptions documentation](#)

CHAPTER 4. MANAGING YOUR CLUSTERS

Use Red Hat OpenShift Cluster Manager to manage and update your clusters.

4.1. VIEWING CLUSTER INFORMATION

The OpenShift Cluster Manager **Clusters** list shows details for all clusters in your organization from:

- OpenShift Container Platform
- OpenShift Dedicated
- Azure Red Hat OpenShift (ARO)
- Red Hat OpenShift Service on AWS (ROSA)



NOTE

To show only the clusters you previously created, set **View only my clusters** to on. To show all clusters in your organization again, set the switch to off.

From here, you can select a cluster to review its settings, check usage, solve issues, and perform other management tasks.

Procedure

- Click a cluster from the list to view more details about it, including:
 - The **Overview** page shows resource usage and basic facts about the cluster
 - The cluster history shows what has happened on this cluster: for example, when it was registered and subscribed to a Red Hat subscription
 - The **Monitoring** tab shows the health of your OpenShift Container Platform cluster and uses the Telemetry service to report the cluster's status in OpenShift Cluster Manager. The Monitoring area shows critical alerts, for example if a cluster operator is failing. This area also shows resource usage.

Additional resources

- See *Remote health monitoring with connected clusters* in the [OpenShift Container Platform documentation](#) for more about monitoring your clusters in OpenShift Cluster Manager.

4.1.1. Determining your cluster ID

Every OpenShift cluster is assigned an ID (in the form of a UUID) when created, but each cluster also has an internal cluster identifier used by OpenShift Cluster Manager. The internal cluster identifier can be changed to a human-readable name displayed in OpenShift Cluster Manager if desired.

You can find this information in OpenShift Cluster Manager, using the command line, or in the OpenShift web console.

Additionally, when OpenShift Container Platform clusters register to OpenShift Cluster Manager, the only identifying information may be the cluster UUID. If multiple OpenShift Container Platform clusters have been registered at the same time, it might be necessary to use the cluster UUID to tell them apart.

Prerequisites

- A Red Hat login
- A Red Hat OpenShift cluster

Procedure

1. Your clusters are listed by ID in [OpenShift Cluster Manager](#) in the **Clusters** area. From here, you can search for a cluster by name or ID. You can also filter your search by cluster type: OpenShift Container Platform (OCP), OpenShift Dedicated (OSD), or Red Hat OpenShift Service on AWS (ROSA).



NOTE

To rename your cluster to a human-readable name, see [Section 4.3, “Renaming your cluster”](#).

- You can also get your OpenShift cluster ID by running the following command locally or on the cluster itself (after logging into the cluster using **oc login**):

```
$ oc get clusterversion <version> -o jsonpath='{.spec.clusterID}{"\n"}'
```

1. You can also find your OpenShift cluster ID in the OpenShift Container Platform web console if you are logged in as an administrator:
 - a. In the details pane on the **Home/Dashboards** page
 - b. On the **Administration/Cluster Settings** page



NOTE

You cannot log in to your Red Hat account with your email unless your email is your Red Hat login. For more information, see [Finding your login](#).



NOTE

You cannot log in to your Red Hat account with your email unless your email is your Red Hat login. For more information, see [Finding your login](#).

4.2. UPDATING BILLING ACCOUNTS FOR OPENSIFT SERVICE ON AWS HOSTED CONTROL PLANES CLUSTERS

If you're a Red Hat OpenShift Cluster Manager cluster administrator and you have at least two billing accounts linked to your Red Hat organization, when you provision a Red Hat OpenShift Service on AWS Hosted Control Planes cluster, you can use the ROSA CLI or the OpenShift Cluster Manager UI to check your billing accounts and make account updates.

Procedure

1. Log in to the Red Hat Hybrid Cloud Console.



NOTE

You cannot log in to the Hybrid Cloud Console with your email unless your email is your Red Hat login. For more information, see [Finding your login](#).

2. From the OpenShift navigation menu, select **Cluster List**. OpenShift Cluster Manager opens.
3. From your list of clusters, click the name of the one with the billing account you want to manage. The **Overview** tab opens.
4. Scroll until you see the **AWS billing account** field. Click the account number under that field. The **Edit AWS billing account** modal opens.
5. Click **Update**.

4.3. RENAMING YOUR CLUSTER

You can give your connected cluster a human-readable name rather than a cluster UUID to make it easier to reference when contacting Red Hat Support or opening a support case, or when reviewing the list of clusters in OpenShift Cluster Manager.

When created, every OpenShift cluster is assigned a 36-character UUID string as a name to differentiate it from other clusters. However, as the UUID can be difficult to search or reference, Red Hat recommends providing a custom name for the cluster to simplify locating resources and managing your OpenShift environment.

Organization Administrators can edit the display name of any cluster within their organization.

Prerequisites

- A Red Hat login
- A Red Hat OpenShift cluster
- You must have the Cluster Owner or Cluster Editor role on the cluster, or Organization Administrator privileges in your Red Hat account to change a cluster's display name in OpenShift Cluster Manager.

Procedure

1. Go to the **Clusters** list in [OpenShift Cluster Manager](#).
2. Click  (**more options**) next to the cluster you want to rename.
3. Click **Edit display name** and enter a name for the cluster.
4. Click **Edit** to save the new name.



NOTE

You can also rename a cluster from its details page from the **Actions** menu > **Edit display name**.

The new cluster name shows in the clusters list on OpenShift Cluster Manager.

Additional resources

- [Section 4.1.1, “Determining your cluster ID”](#)

4.4. TAGGING CLUSTERS AS HUB CLUSTERS

You can tag a cluster in Red Hat OpenShift Cluster Manager as a Red Hat Advanced Cluster Management for Kubernetes (RHACM) hub cluster. By tagging your cluster, you identify it as a resource that uses RHACM to manage other clusters in your fleet. This tag also enables OpenShift Cluster Manager to categorize the cluster as a hub, which centralizes management and visibility for your registered resources across the Red Hat Hybrid Cloud Console.



NOTE

This tag only adds the cluster to the hub clusters list and is for organization purposes only. Tagging a cluster as a hub cluster does not install RHACM on the cluster. You must have RHACM installed on the cluster before you can tag it.

Prerequisites

- You’re logged in to OpenShift Cluster Manager.
- You have a cluster that is already installed and registered.
- You have installed RHACM on the cluster you want to tag.
- You have administrator permissions for the cluster in OpenShift Cluster Manager.

Procedure

1. From the navigation menu on the OpenShift **Overview** page, select **Fleet management > Clusters**.
2. Select the name of the cluster you want to tag. You can select more than one cluster to tag multiple clusters at the same time.
3. Click **Add cluster**, then select **Tag as hub cluster**.
4. Click **Tag** to confirm.



NOTE

You can remove a tag by selecting the cluster and clicking **Remove hub cluster tag** in the overflow menu. You can also select multiple clusters and click **Remove hub cluster tag**. Click **Remove tag** to confirm.

Verification

- From your **Clusters** list, make sure you see the **RHACM hub cluster** label under the name of the cluster you tagged.

Additional resources

- [Learn more about RHACM installation](#)
- [Learn more about the RHACM multicluster global hub](#)

4.5. DOWNLOADING AND UPDATING PULL SECRETS

You can manage your cluster pull secrets by downloading them from the console or updating them globally.

4.5.1. Downloading the pull secret from OpenShift Cluster Manager

An image pull secret provides authentication for the cluster to access services and registries which serve the container images for OpenShift components. Every individual user gets a single pull secret generated.

The pull secret is used when installing an OpenShift Container Platform cluster. It is also used by OpenShift Cluster Manager to identify a specific Red Hat user when transferring cluster ownership. To transfer a cluster to another owner, you need to download the pull secret for the user that will take ownership of the cluster.

Prerequisites

- A Red Hat login

Procedure

1. Log in to [OpenShift Cluster Manager](#) as the Red Hat user you want to download the pull secret for.



IMPORTANT

Each pull secret is unique to a specific user. If you are downloading the pull secret to transfer a cluster to another owner, you must log in to OpenShift Cluster Manager as the user that will take ownership of the cluster, and obtain that user's pull secret.

2. Go to [Downloads](#) in OpenShift Cluster Manager and find your pull secret in the **Tokens** category.
 - Click **Copy** to copy your pull secret to the clipboard.
 - Click **Download** to download your pull secret.



IMPORTANT

Do not share your pull secret. The pull secret should be treated like a password.

You can now use this pull secret to create an OpenShift Container Platform cluster or for transferring cluster ownership.

Additional resources

- To transfer a cluster to a different owner, see [Transferring cluster ownership](#).

- To create an OpenShift Container Platform cluster, see the [OpenShift Container Platform installation documentation](#).
- For more details about using pull secrets, see [Using image pull secrets](#) in the OpenShift Container Platform documentation.

4.5.2. Updating the global pull secret

An image pull secret provides authentication for the cluster to access services and registries which serve the container images for OpenShift components. Every individual user gets a single pull secret generated.

The pull secret is used when installing an OpenShift Container Platform cluster and when transferring cluster ownership.

To transfer a connected cluster to a new owner, you must update the pull secret on a cluster to the new owner's pull secret after initiating a cluster transfer in OpenShift Cluster Manager. The pull secret must be updated within five days of initiating the transfer process, or the process will need to be initiated again from OpenShift Cluster Manager. See [Section 4.6, "Cluster ownership transfer"](#).



IMPORTANT

On clusters using OpenShift Container Platform versions earlier than 4.7.4, cluster resources must adjust to the new pull secret. This can temporarily limit the usability of the cluster. This occurs because updating the pull secret causes the Machine Config Operator to drain the nodes, apply the change, and uncordon the nodes.

This does not affect clusters using OpenShift Container Platform versions 4.7.4 and later, where a pull secret change does not cause a node drain or reboot.

Prerequisites

- An OpenShift Container Platform cluster
- A Red Hat login with Cluster Owner or Organization Administrator privileges on the cluster in OpenShift Cluster Manager
- A new or modified pull secret file to upload. You can download your pull secret from [Downloads](#) from the **Tokens** area.
- You have access to the cluster as a user with the **cluster-admin** role. See [Authentication and authorization](#) in the OpenShift Container Platform documentation for more information about cluster roles.
- If you are transferring the cluster to a new owner, you must initiate the transfer in OpenShift Cluster Manager *before* changing the global pull secret to be able to receive Telemetry metrics to monitor the cluster.

Procedure

1. Run the following command using the pull secret you downloaded from OpenShift Cluster Manager to change the cluster's pull secret:

```
# oc set data secret/pull-secret -n openshift-config --from-file=.dockerconfigjson=pull-secret.txt
```

If a secret is not already created, run the following command to create the secret:

```
# oc create secret generic pull-secret -n openshift-config --
type=kubernetes.io/dockerconfigjson --from-file=.dockerconfigjson=/path/to/downloaded/pull-
secret
```

2. Wait for the update process to complete on all nodes in the cluster. The time required depends on the size of your cluster.
3. Verify the pull secret change by going to [Downloads](#) in OpenShift Cluster Manager and finding your pull secret in the **Tokens** category:
 - a. Click **Copy** to copy your pull secret to the clipboard.
 - b. Click **Download** to download your pull secret.

Additional resources

- [Using image pull secrets](#)
- [Transferring cluster ownership](#)

4.5.3. Creating a pull secret

You can create a new pull secret in certain cases. For example, if your current pull secret is not working, or if you need a new pull secret for security reasons.

To create a pull secret, follow the instructions in the KCS article [How to request pull-secret rotation](#).



NOTE

Using this method requires use of the **ocm-cli** tool. For more information about the **ocm-cli** tool, see [Using the ocm-cli to manage clusters in OpenShift Cluster Manager](#).

4.6. CLUSTER OWNERSHIP TRANSFER

You can transfer ownership of an OpenShift Container Platform cluster to another user in your organization or a different organization using OpenShift Cluster Manager.

For example, if you created an OpenShift Container Platform cluster using one Red Hat account but want to move the cluster to a different Red Hat account to register it to the associated subscription, you need to transfer cluster ownership to that user. You can transfer ownership of connected or disconnected clusters.



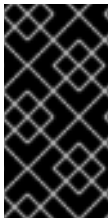
NOTE

To transfer ownership of an OpenShift Dedicated or hosted Red Hat OpenShift Service on AWS (ROSA) cluster to another user, open a [customer support case](#) with Red Hat Support. To transfer a ROSA Classic cluster, see [Section 4.6.3, "Initiating ownership transfer of a ROSA Classic cluster"](#).

Connected clusters

Transferring ownership of a connected cluster requires two steps: initiate the transfer in OpenShift Cluster Manager, then change the cluster's pull secret from the command line. You must change the cluster pull secret within five days of initiating the transfer, or you need to restart the transfer procedure.

The transfer is complete when OpenShift Cluster Manager begins receiving Telemetry data from the cluster with the new pull secret. See [Transferring ownership of a connected cluster](#) for instructions.



IMPORTANT

The cluster transfer does not complete successfully if only the pull secret is updated to the new cluster owner. As a result, the cluster might stop reporting Telemetry metrics for monitoring. You must initiate the ownership transfer in OpenShift Cluster Manager in addition to changing the cluster pull secret to complete the transfer.

Disconnected clusters

To transfer ownership of a disconnected cluster, you only need to initiate the transfer in OpenShift Cluster Manager; no pull secret update is required. The transfer is complete when the new cluster owner registers the cluster to OpenShift Cluster Manager. See [Transferring ownership of a disconnected cluster](#) for instructions.

4.6.1. Transferring ownership of a connected cluster

You can transfer ownership of a connected OpenShift Container Platform cluster to another user in your organization or a different organization using OpenShift Cluster Manager.



NOTE

This procedure outlines transferring connected clusters. To transfer a disconnected cluster, see [Section 4.6.2, "Transferring ownership of a disconnected cluster"](#).

To transfer a connected cluster to another owner, you must:

1. Initiate the transfer in OpenShift Cluster Manager.
2. Change the cluster pull secret to the new owner's pull secret from the command line within five days of initiating the transfer.

Prerequisites

- A Red Hat login.
- An OpenShift Container Platform cluster.
- You must be the Cluster Owner on the cluster, or an Organization Administrator in the associated Red Hat account. See [User access concepts in OpenShift Cluster Manager](#) for more information.



NOTE

To create a new user to take over cluster ownership, see [How to Create and Manage Users](#) on the Red Hat Customer Portal.

Procedure

1. Log into [OpenShift Cluster Manager](#) as the current cluster owner.
2. Initiate the transfer:
 - a. Select the cluster that you want to transfer from the **Clusters** list.
 - b. Click **Actions** > **Transfer cluster ownership** at the top of the cluster's details page.
 - c. Click **Initiate transfer** to confirm this action.



IMPORTANT

1. You must change the cluster's pull secret within five days of initiating the transfer and register the cluster with the new Red Hat account or the transfer is cancelled.
2. You can cancel the ownership transfer anytime before the pull secret has been changed by clicking **Actions** > **Cancel ownership transfer**.

You have now initiated the ownership transfer. The next step is to change the cluster's pull secret to the pull secret of the new cluster owner.

Verification

Log into [OpenShift Cluster Manager](#) as the new owner of the cluster. You can verify the transfer was successful by checking these details in the cluster **Overview**:

- In **Details**, the **Owner** has been updated.
- In **Cluster history**, details of the transfer appear.

If the cluster was transferred to a different organization, you can log into that organization to verify the update. The cluster now appears in the target Red Hat account's clusters list, and has been removed from the previous Red Hat account clusters list.

The transfer is complete when OpenShift Cluster Manager receives Telemetry data from the cluster with the new pull secret.

4.6.2. Transferring ownership of a disconnected cluster

You can transfer ownership of a disconnected OpenShift Container Platform cluster to another user in your organization or a different organization using OpenShift Cluster Manager.

To transfer ownership of a disconnected cluster, you only need to initiate the transfer in OpenShift Cluster Manager.

Prerequisites

- A Red Hat login
- An OpenShift Container Platform cluster
- You must be the Cluster Owner on the cluster, or an Organization Administrator in the associated Red Hat account. See [User access concepts in OpenShift Cluster Manager](#) for more information.

Procedure

1. Log into [OpenShift Cluster Manager](#) as the current cluster owner.
2. Initiate the transfer:
 - a. Select the cluster that you want to transfer from the **Clusters** list.
 - b. Click **Actions** > **Transfer cluster ownership** at the top of the cluster's details page.
 - c. Click **Initiate transfer** to confirm this action.
3. Provide the cluster UUID to the user that you are transferring the cluster to.



NOTE

You can find the cluster UUID on the cluster details page in OpenShift Cluster Manager (Cluster ID), or on the **About** page of the cluster web console in OpenShift Container Platform.

4. As the new cluster owner, log into OpenShift Cluster Manager.
5. Register the disconnected cluster with the cluster UUID using the steps in [Registering disconnected clusters](#).

When the cluster registers to OpenShift Cluster Manager successfully, the cluster ownership transfer is complete.

Verification

Log into [OpenShift Cluster Manager](#) as the new owner of the cluster. You can verify the transfer was successful by checking these details in the cluster **Overview**:

- In **Details**, the **Owner** has been updated.
- In **Cluster history**, details of the transfer appear.

If the cluster was transferred to a different organization, you can log into that organization to verify the update. The cluster now appears in the target Red Hat account's clusters list, and has been removed from the previous Red Hat account clusters list.

It is recommended to update the pull secret to match the new owner once the transfer is complete. For more information about updating the pull secret, see [Updating the global pull secret when transferring cluster ownership](#).

4.6.3. Initiating ownership transfer of a ROSA Classic cluster

You can use OpenShift Cluster Manager to transfer ownership of a ROSA Classic cluster to another user in your organization or a different Red Hat organization.

Prerequisites

- A Red Hat login.
- A ROSA Classic cluster.

- You are the Cluster Owner of a cluster that you want to transfer, or an Organization Administrator in the associated Red Hat account. See [User access concepts in OpenShift Cluster Manager](#) for more information.



NOTE

To create a new user to take over cluster ownership, see [How to Create and Manage Users](#) on the Red Hat Customer Portal.

Procedure

1. Log in to [OpenShift Cluster Manager](#).



NOTE

You cannot log in to your Red Hat account with your email unless your email is your Red Hat login. For more information, see [Finding your login](#).

2. Click **Cluster List** and locate the cluster that you want to transfer.
3. Click the options icon () on the far right of the cluster row and select **Transfer Cluster**. The **Transfer ownership** window opens.
4. Enter the username of the new cluster owner, the account ID, and the organization ID, and then click **Initiate transfer**. The user that you enter will receive an email request to approve the ownership transfer. The transfer request expires after 15 days.
5. To view your cluster transfer requests, select **Cluster List** from the left navigation panel and then click **View cluster requests**. The **Transfer Ownership Request** window opens.



NOTE

To cancel the transfer request, click **Cancel** next to the transfer that you want to cancel.

4.6.4. Accepting ownership transfer of a ROSA Classic cluster

A ROSA Classic cluster owner can initiate the transfer of a ROSA Classic cluster to you. After the transfer has been initiated, you must accept the transfer in OpenShift Cluster Manager to complete the transfer.

Prerequisites

- A Red Hat login.
- A ROSA Classic cluster.
- You are a ROSA Classic cluster owner, or an Organization Administrator in the associated Red Hat account. See [User access concepts in OpenShift Cluster Manager](#) for more information.

Procedure

1. Log in to [OpenShift Cluster Manager](#).

**NOTE**

You cannot log in to your Red Hat account with your email unless your email is your Red Hat login. For more information, see [Finding your login](#).

2. Select **Cluster List** from the left navigation panel and then click **View cluster requests**. The **Transfer Ownership Request** window opens.
3. For the cluster that you want to transfer, click **Accept/Decline**. The **Complete cluster transfer** window opens.
4. Click **Approve Transfer**. The transfer can take up to 24 hours to complete.

Verification

- Verify that the cluster transfer was successful:
 - In **Details**, the **Owner** has been updated.
 - In **Cluster history**, details of the transfer display.
 - If the cluster was transferred to a different organization, log in to that organization to verify the update. The cluster now appears in the target Red Hat account clusters list, and has been removed from the previous Red Hat account clusters list.

4.7. MONITORING CLUSTERS

You can monitor the status of your clusters, use Red Hat Lightspeed Advisor to manage your cluster health, troubleshoot a stale cluster, and create integrations with third-party applications to receive cluster event notifications.

4.7.1. Configuring cluster notifications with third-party applications

In addition to email notifications, you can receive your cluster event notifications through third-party applications that are integrated with your Red Hat account.

Prerequisites

- You have a Red Hat account with one of the following roles:
 - Organization Administrator
 - Cloud Administrator
 - Notifications Administrator

Procedure

1. In the [Hybrid Cloud Console](#), click the **Settings** icon (⚙) and then select **Integrations**.

**NOTE**

You cannot log in to the Hybrid Cloud Console with your email unless your email is your Red Hat login. For more information, see [Finding your login](#).

2. Select the **Communications** or **Reporting & Automation** tab, and then click **Add integration**.
3. Select the third-party application where you would like to receive notifications, and then follow the instructions in the integration wizard.

Additional resources

For more information about Hybrid Cloud Console notifications integrations, see [Integrating the Red Hat Hybrid Cloud Console with third-party applications](#).

4.7.2. Monitoring cluster status

With the Red Hat OpenShift Cluster Manager, you can manage your clusters and monitor them for potential issues. The **Status** column in the **Clusters** list reports each cluster's state so you can monitor your clusters at a glance.

The cluster status is shown in the **Clusters** list in [OpenShift Cluster Manager](#). Normally, a cluster shows a status of **Ready**, however, there are other possible cluster statuses.

Table 4.1. Cluster status

Cluster status	Description
Waiting	STS clusters only. Waiting for users to complete the necessary tasks before the cluster can be installed.
Validating	CCS clusters only. Waiting for users to complete the necessary tasks before the cluster can be installed.
Pending	Interim state after users complete the required tasks but before the cluster installation starts.
Installing	The cluster is currently being installed.
Ready	Cluster is installed and is ready for use. Cluster is connected to Red Hat OpenShift Cluster Manager and is reporting Telemetry data. No manual registration is required.
Error	Failure in creating or destroying a cluster.
Disconnected	Cluster does not send Telemetry data to OpenShift Cluster Manager. The disconnection could be because the cluster is installed on a private network, or because the Telemetry is disabled.

Cluster status	Description
Stale	The cluster is connected but has not send Telemetry data to OpenShift Cluster Manager recently. You can no longer view internal information about the cluster in the OpenShift Cluster Manager Monitoring tab, including any alerts firing, cluster operators, and resource usage. To know the workaround steps, see Troubleshooting a stale cluster.
Limited support	When a cluster transitions to a Limited Support status, it means that the SLA is no longer applicable and credits requested against the SLA are denied. It does not mean that you no longer have product support. The cluster can return to a fully-supported status if the violating factors are remediated. For more information, see Limited support status.
Powering down	Cluster goes into hibernation mode.
Hibernating	Cluster is inactive. Users can choose to move clusters into hibernation to reduce maintenance costs.
Resuming	Interim state after users choose to take clusters from hibernation mode. Cluster is powering on but not fully functional yet.
Uninstalling	Cluster is being uninstalled.
Archived	Clusters are moved from the cluster list view to the Archive list. In the archived list, you can see two tabs only: Overview and Support .

4.7.3. Using Insights Advisor recommendations to manage cluster health

Red Hat Insights Advisor for OpenShift Container Platform allows you to assess and monitor the health of your OpenShift Container Platform clusters from the Red Hat Hybrid Cloud Console.

Insights Advisor highlights service availability, fault tolerance, performance, and security risks for your OpenShift Container Platform clusters based on Red Hat recommendations, so that you can avoid potential problems or solve problems quickly without unplanned downtime.

Recommendations include information about detected issues, including risk level, affected clusters, and steps for resolution where applicable.

To use Insights Advisor, your cluster must be registered to OpenShift Cluster Manager. To register a disconnected cluster, see [Registering OpenShift Container Platform clusters to OpenShift Cluster Manager](#).

Additional resources

- See [Remote health monitoring with connected clusters](#) and [Monitoring your OpenShift cluster health with Insights Advisor](#) for information about using Insights Advisor for OpenShift Container Platform.
- See [Red Hat Insights Data & Application Security](#) for information about Red Hat Insights data collection and controls.

4.7.4. Troubleshooting a stale cluster

OpenShift Cluster Manager allows you to manage your clusters and monitor them for potential issues. The **Status** column in the **Clusters** list reports each cluster's state so you can monitor your clusters at a glance.

An OpenShift Container Platform cluster can be shown as **Stale** in OpenShift Cluster Manager even if it is running and functioning normally. However, when a cluster status becomes **Stale**, you can no longer view internal information about the cluster in the OpenShift Cluster Manager **Monitoring** tab, including any alerts firing, cluster operators and resource usage.

In most cases, a cluster reports a **Stale** status because the **telemeter-client** cannot contact the Red Hat Telemetry endpoints. Telemetry enables the cluster to communicate with OpenShift Cluster Manager. If a cluster is reporting a **Stale** status, you can restore it to **Ready** and resume monitoring cluster health with the steps in this section.

Prerequisites

- A Red Hat login
- An OpenShift Container Platform cluster
- You have access to the cluster as a user with the **cluster-admin** role.
- You have installed the OpenShift CLI (**oc**).

Procedure

To diagnose a stale cluster and restore it to **Ready** status:

1. Check that the OpenShift Container Platform cluster can connect to the Telemetry endpoints. If applicable, ensure that the cluster proxy configuration allows access to the Telemetry endpoints. The required Telemetry endpoints can be found in the [Configuring your firewall for OpenShift Container Platform](#) in the OpenShift Container Platform *Installing* documentation.
2. Review the logs of the **telemeter-client** pod running in the **openshift-monitoring** namespace for any errors:

```
$ oc get pods -n openshift-monitoring -l app.kubernetes.io/name=telemeter-client
```

3. Restart the **telemeter-client** pod in the **openshift-monitoring** namespace and the **insights-operator** pod in the **openshift-insights** namespace:

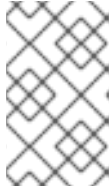
```
$ oc delete pod -n openshift-monitoring -l app.kubernetes.io/name=telemeter-client
$ oc delete pod -n openshift-insights -l app=insights-operator
```

- Wait a few minutes after restarting the pods, then locate your cluster in the **Clusters** list in [OpenShift Cluster Manager](#) and check that the **Status** column shows **Ready**.

After completing these steps, if you can view the cluster in OpenShift Cluster Manager but the status is still **Stale**, occasionally this may be because the cluster is owned by an inactive user in your Red Hat organization. If this is the case, you can restore the cluster to **Ready** by transferring cluster ownership to an active user in your organization.

- To determine the cluster owner, check the cluster pull secret using the following command:

```
$ oc get secret pull-secret -n openshift-config -o jsonpath='{.data.\.dockerconfigjson}' |
base64 -d | jq
```



NOTE

The **jq** JSON processor is needed to execute the above command and make the output more readable. You can also run the above command without using **jq** by removing **| jq** from the command, but the output is more difficult to read.

- Check that the user account in the secret is still active in the [Red Hat Customer Portal](#).
- If the user is no longer active, transfer your cluster to another owner using the steps in [Transferring cluster ownership](#). When the ownership transfer is complete, the cluster will show in OpenShift Cluster Manager as **Ready**.

Verification steps

- Locate your cluster in the **Clusters** list in [OpenShift Cluster Manager](#) and check that the **Status** column shows **Ready**. In a few minutes, you will also see information appear about the cluster in the OpenShift Cluster Manager **Monitoring** tab.

Additional resources

- [Verifying your OpenShift Container Platform cluster is registered and subscribed](#)
- [Red Hat Insights for OpenShift Container Platform \(*Remote health monitoring with connected clusters*\) documentation](#)
- [OpenShift Container Platform documentation](#)

4.8. DOWNLOADING COMMAND LINE (CLI) TOOLS

The [Downloads](#) page in OpenShift Cluster Manager provides a single place to download CLI tools and find your authentication tokens to manage OpenShift.

The **Downloads** page includes command line tools such as:

- Command-line interface (CLI) tools to manage and work with OpenShift from your terminal
- Developer tools to simplify the use of Kubernetes
- OpenShift installers to create OpenShift Container Platform and CodeReady Containers clusters.

- Red Hat Enterprise Linux CoreOS (RHCOS) management tools for customizing your RHCOS nodes.
- Tokens for authentication, including your pull secret and OpenShift Cluster Manager API token.

Procedure

1. Go to [Downloads](#) and find the resource you want to download.
2. (Optional) Expand the tool or token description to learn more about the download and see links to related documentation.
3. Specify the operating system and architecture you are using in the **OS type** and **Architecture type** dropdowns, and click **Download**.

Additional resources

- [Downloading and updating pull secrets](#)

4.9. DOWNLOADING THE OPENSIFT CLUSTER MANAGER API TOKEN

Use your OpenShift Cluster Manager API token to authenticate against your OpenShift Cluster Manager account.

The API token is required to connect to OpenShift Cluster Manager to use the **rosa CLI** and **ocm-cli** command line tools. You can use the same token with both services.

For security, tokens are hidden from display in OpenShift Cluster Manager by default. You can access your API token on the OpenShift Cluster Manager [Downloads](#) page, then view or copy it to use in the command line.

- For information about using the **rosa CLI**, see the [rosa CLI documentation](#).
- For more information about **ocm-cli**, see [Using the ocm-cli to manage your clusters in OpenShift Cluster Manager](#).



NOTE

The **ocm-cli** tool is currently Development Preview.

A release that is provided as Development Preview is provided to a limited set of customers for their evaluation of an early version of the product and collection of feedback back to the product development teams. Development Preview releases are not supported in production environments.

Prerequisites

- A Red Hat login

Procedure

1. Go to [Downloads](#) and find the **OpenShift Cluster Manager API Token** row under **Tokens**.

2. Click **View API token** to go to the **OpenShift Cluster Manager API Token** page.
3. Click **Load token** to display your token. By default, the token is hidden from display.
4. Click  (**Copy to clipboard**) to copy your token to use in a terminal.

Additional resources

- See [Using the ocm-cli to manage your clusters in OpenShift Cluster Manager](#) .

4.9.1. Revoking OpenShift Cluster Manager API tokens

You can revoke offline tokens in case you no longer need them or if you have reached the token limit.

Procedure

1. Navigate to the [SSO application management](#) page.
2. Locate the **cloud-services** client ID; expand the row if necessary.
3. Select **Remove access**.
All refresh tokens will stop working immediately after you remove access, but existing access tokens (which are cached by **ocm** and **rosa** commands) might take up to 15 minutes to expire.

Refresh the page afterwards to generate a new token.

4.10. AUTHENTICATING THE OPENSIFT CLUSTER MANAGER CLI WITH RED HAT SINGLE SIGN-ON

Use Red Hat Single Sign-On (SSO) to securely authenticate with OpenShift Cluster Manager. Single sign-on authentication provides a more secure alternative to long-lived offline tokens, which pose significant security risks.

Offline authentication tokens are long-lived and difficult to revoke, which increases the window of opportunity for unauthorized access. Furthermore, because these tokens are often stored locally on your machine, other applications can potentially access them if they are not properly secured. The manual process of copying and pasting these tokens also introduces unnecessary risk. In contrast, SSO authentication provides a more secure and streamlined experience by using temporary authorization codes that are unique to each session. This method significantly reduces the risk of unauthorized access and enhances the overall security of your interactions with OpenShift Cluster Manager.

To ensure security, Red Hat recommends using the **ocm-cli** command-line tool with Red Hat SSO instead of an offline token. Single sign-on authentication is supported with the OpenShift Cluster Manager CLI version v0.1.73 or later.



NOTE

The **ocm-cli** tool is currently Developer Preview.

A Developer Preview release is provided to a limited set of customers for their evaluation of an early version of the product and collection of feedback back to the product development teams. Developer Preview releases are not supported in production environments. For more information, see [Developer Preview Support Scope](#).

While you can still log in to the Red Hat **ocm-cli** tool with an offline token, you can log in more securely with the Red Hat secure browser-based single sign-on (SSO) method, which automatically sends your CLI instance a refresh token that is valid for 10 hours. Because this authorization code is unique and temporary, it is a secure method of authentication. An SSO authorization code gives you access to any Red Hat OpenShift services you need with just one sign-on.

4.10.1. Next steps

- If your system has a web browser, follow the steps in [Section 4.10.2, “Logging in to the OpenShift Cluster Manager CLI tool with a Red Hat Single Sign-On authorization code”](#) to authenticate with Red Hat Single Sign-On.
- If you are working with containers, remote hosts, or other environments without a web browser, follow the steps in [Section 4.10.3, “Logging in to the OpenShift Cluster Manager CLI with a Red Hat Single Sign-On device code”](#) to authenticate with Red Hat Single Sign-On.

Additional resources

- [Using the ocm-cli to manage clusters in OpenShift Cluster Manager](#)

4.10.2. Logging in to the OpenShift Cluster Manager CLI tool with a Red Hat Single Sign-On authorization code

Log in to the OpenShift Cluster Manager CLI using a Red Hat Single Sign-On (RH-SSO) authorization code. This authentication method, which requires a web-based browser, provides a more secure alternative to long-lived offline tokens.

Prerequisites

- You have a Red Hat account.
- You have the OpenShift Cluster Manager CLI (ocm) version v0.1.73 or later.
- Your system has a supported web browser installed. For information about supported browsers, click **Browser support** at the bottom of the [Hybrid Cloud Console](#).

Procedure

1. In a terminal window, enter the following command:

```
ocm login --use-auth-code
```

A Red Hat SSO authentication browser window opens.

2. Log in with your Red Hat account credentials.



NOTE

You cannot log in to your Red Hat account with your email unless your email is your Red Hat login. For more information, see [Finding your login](#).

3. After the authentication process completes, close the login window and go back to the CLI in the terminal window. The CLI displays the message **Login Successful**. You can now use the **ocm-cli** tool to interact with all OpenShift Cluster Manager APIs.

4.10.3. Logging in to the OpenShift Cluster Manager CLI with a Red Hat Single Sign-On device code

If you're working with containers, remote hosts, and other environments without a web browser, you can use a Red Hat Single Sign-On device code for secure authentication. To do this, use a second device that has a web browser to approve the login.



NOTE

This method is more secure than using an offline token.

Prerequisites

- You have a Red Hat account.
- You have the OpenShift Cluster Manager CLI (ocm) version v0.1.73 or later.
- You have a second device that has a web browser.

Procedure

1. In a terminal window on the device that you want to authenticate, enter the following command:

```
ocm login --use-device-code
```

The CLI displays a message similar to the following message where **<DEVICE_CODE>** is the unique authentication code:

```
To login, navigate to https://sso.redhat.com/device on another device and enter code  
<DEVICE_CODE>. Checking status every 5 seconds...
```

2. On the device that has a web browser, navigate to <https://sso.redhat.com/device>.
3. In the **Device log in** window, enter the device code from the CLI message, and then click **Submit**.
4. In the **Grant access to ocm-cli** window, click **Grant access**. Your browser displays the message **Red Hat device login successful**. You can now use the **ocm-cli** tool to interact with all OpenShift Cluster Manager APIs.

Verification

Return to the terminal on the device that you authenticated and confirm that the CLI displays this message:

```
Login successful.
```

4.10.4. Offline tokens and automation

The new secure method of authenticating using Red Hat Single Sign-On will not break any existing automations that rely on offline tokens. To use offline tokens for automation or other purposes, you can download the [OpenShift Cluster Manager API Token](#) page. Use service accounts, available on the [Service Accounts](#) page, for automation purposes.

Additional resources

- [Creating and Managing Service Accounts](#)

4.11. MANAGING YOUR ADD-ON SERVICES

From OpenShift Cluster Manager, you can manage the add-ons installed on your OpenShift cloud services clusters. Add-ons are services that you can install to enhance the capabilities of your Red Hat OpenShift Dedicated and Red Hat OpenShift Service on AWS (ROSA) clusters.

To access your add-ons and find information about them, navigate to your cluster's **Add-ons** tab in [OpenShift Cluster Manager](#), and select the add-on.

Additional resources

- To add a service to your OpenShift Dedicated cluster or manage your existing add-ons, see [Add-on Services](#) in the OpenShift Dedicated documentation.
- To add a service to your Red Hat OpenShift Service on AWS (ROSA) cluster or manage your existing add-ons, see [Add-on Services](#) in the ROSA documentation.

CHAPTER 5. CONFIGURING ACCESS TO CLUSTERS IN OPENSIFT CLUSTER MANAGER

OpenShift Cluster Manager allows you to view and manage the OpenShift clusters in your organization from one dashboard.

Viewing and editing access to clusters in OpenShift Cluster Manager is controlled by your Red Hat account configuration (generally by *organization*) and by role bindings configured in OpenShift Cluster Manager.

Your role in your organization, as well as the roles you have been assigned on a cluster, determine how you can manage a cluster, for example:

- Viewing the list of clusters in your organization, including your cluster and clusters created by other users
- Viewing a cluster's details, such as the cluster overview, subscription settings, history, and Cluster Owner
- Editing a cluster's details, such as subscription settings, cluster display name, machine pools, and add-on services

Any user with a Red Hat login has permission to create a cluster from [OpenShift Cluster Manager](#). However, your organization must have sufficient subscriptions or quota, depending on the type of OpenShift cluster you are creating, to allow you to create a cluster. See [Cluster subscriptions and registration](#) for more information about subscriptions and quota for clusters.

When you create a cluster, you are assigned the Cluster Owner role on that cluster.



NOTE

For greater security, you can use two-factor authentication (2FA) to access OpenShift Cluster Manager and the Red Hat Hybrid Cloud Console. To learn more about configuring two-factor authentication, see [Using OpenShift Cluster Manager with the Red Hat Hybrid Cloud Console](#) and the [Using Two-Factor Authentication](#) guide.

5.1. USER ACCESS CONCEPTS IN OPENSIFT CLUSTER MANAGER

Organization

An *organization* is defined in your Red Hat account. An organization can have many users, who each have a login to access Red Hat resources such as the [Red Hat Hybrid Cloud Console](#) and the Red Hat Customer Portal.

In OpenShift Cluster Manager, users can view all clusters created within their organization by default.

Organization Administrator

Each organization has one or more *Organization Administrator* users.

This is the highest permission level in an organization, and the only role that can manage user access and permissions within a Red Hat account. Organization Administrators can access and edit any cluster in the organization, as well as configure user roles on clusters in OpenShift Cluster Manager.

For more information about Red Hat account roles, see [Roles and Permissions for Red Hat Customer Portal](#) and [How To Create and Manage Users](#).

Cluster Owner

The user that creates an OpenShift cluster is the *Cluster Owner*. This user can perform any action on the cluster and view all details about the cluster in OpenShift Cluster Manager.

Cluster Owners can allow other users in the same organization to manage and perform actions on their cluster by granting them the *Cluster Editor* role.

Organization Administrators have the same access to clusters as Cluster Owners.

You can also become the Cluster Owner on an existing cluster when another user transfers a cluster's ownership to you. See [Transferring cluster ownership](#) for more information.

Cluster Editor

The *Cluster Editor* role allows you to edit, manage, and delete that cluster, similar to Cluster Owner. The one exception is that a Cluster Editor cannot grant roles on a cluster to other users. Only a Cluster Owner or an Organization Administrator in the Red Hat account can configure role bindings on clusters.

5.2. CONFIGURING USER ACCESS TO CLUSTERS IN OPENSIFT CLUSTER MANAGER

You can configure user access to your clusters by assigning specific roles to users in OpenShift Cluster Manager.

5.2.1. Viewing user roles and access on a cluster

You can view a list of users with assigned roles on a cluster from the **OCM Roles and Access** screen.

If you are an Organization Administrator in the Red Hat account or the Cluster Owner, you can also edit the users and their access to the cluster from this screen. Other users can only view information about users and roles on a cluster.

Prerequisites

- A Red Hat login
- An existing OpenShift cluster in your organization

Procedure

1. Select your cluster from the **Clusters** list.
2. Click **Access Control** > **OCM Roles and Access** to see a list of users with assigned roles to access the cluster.

5.2.2. Granting user roles for cluster access

After you create an OpenShift cluster, you can grant access to other users on your cluster. This enables members of your team to manage or view the cluster without being an Organization Administrator in the Red Hat account.

Prerequisites

- A Red Hat login.
- An existing OpenShift cluster.
- You must be the Cluster Owner on the cluster, or Organization Administrator in your Red Hat account
- The user you want to grant access to must be in your organization

Procedure

1. Select your cluster from the **Clusters** list.
2. Click **Access Control > OCM Roles and Access**
3. Click **Grant role**.
4. Enter the Red Hat login for the user.
5. Select the role you want (for example, Cluster Viewer) from the list.
6. Click **Grant role** to confirm the role assignment.

Verification

The user is listed on the **OCM Roles and Access** screen with the role assigned.

5.2.3. Revoking user roles for clusters

You can revoke a user's cluster permissions if you are the Cluster Owner or Organization Administrator.

Prerequisites

- A Red Hat login.
- An existing OpenShift cluster.
- You must be the Cluster Owner on the cluster, or Organization Administrator in your Red Hat account
- A user in your organization with access to the cluster

Procedure

1. Select your cluster from the **Clusters** list.
2. Click **Access Control > OCM Roles and Access**
3. Click  (**more options**) next to the user on the list, then **Delete**.
4. Click **Confirm**.

Verification

The user is not displayed in the users list in **OCM Roles and Access**.

5.2.4. Using role-based access control to assign users and groups

You can use role-based access control (RBAC) to create and manage groups of users. Assigning roles to groups allows you to manage access for users as a group. Roles assigned using RBAC apply to all clusters within your organization rather than a specific cluster. RBAC is available in the **Identity & Access Management** menu in the **Settings** gear of the Red Hat Hybrid Cloud Console.



NOTE

Only organization administrators can manage and assign roles to groups using role-based access control (RBAC).

Organization administrators can change the default access permissions for the users within their organization. Role-based access control defaults to two groups. The *default member group* has all users within the organization as members. The *default admin group* has all users with the organization administrator role as members.

OCM access policies are explicitly assigned through role assignments to these default groups. In this way, the existing RBAC policies are no longer explicit and customers can modify them. Organization administrators are able to remove the role assignments from the default groups to remove default permissions from all users. They can then assign these roles selectively to specific users or groups to manage the permissions for users within their organization.



IMPORTANT

Removing all OCM roles from the default groups results in users losing the ability to view and provision clusters. It is recommended to set up groups of users and assign specific roles to these groups before revoking access from the default groups.

A scope governs the level that the role is applied or granted to a user or a group. There are two scopes used within OCM, *cluster scope* and *organization scope*. Roles can be granted to a user or a group at either a cluster scope or at the organization scope.

A role granted at the cluster scope enables the user the ability to take the allowed action (as specified by the permissions included within the role) for the specific cluster that the role is being granted for. Essentially, cluster scoped role assignments are for a specific cluster.

A role granted at the organization scope enables the user the ability to take the allowed action (as specified by the permissions included within the role) for all clusters within the organization. Essentially, organization scoped role assignments are cross-cluster and apply to all clusters within the organization.

Users can create and manage groups and group membership for users within the organization through the RBAC service within the [Red Hat Hybrid Cloud Console](#).

Users can assign a role to a group by using RBAC. Any role assigned with RBAC is at the organization level and applies to all clusters within the organization.

Users can assign a role to a user in OCM for a particular cluster. They do this from within the context of a particular cluster and this role assignment is at the cluster scope.

For more information about using RBAC within Red Hat Hybrid Cloud Console, read [User Access Configuration Guide for Role-based Access Control \(RBAC\)](#).

5.2.4.1. Using RBAC to assign roles and users to groups

When adding roles to created groups, you can add OCM-specific roles. Use these OCM-specific roles to give users or groups in your organization more precise access to clusters. When adding roles inside your group, use the search box and type "OCM" to find all the OCM-specific roles you can add.

The roles you can add are:

- **Cluster viewer:** This role allows a user to view a cluster.
- **Cluster provisioner:** This role allows a user to provision a cluster.
- **Cluster editor:** This role allows a user to manage and delete a cluster.
- **Organization administrator:** This role allows a user to perform all tasks within OCM for all clusters. Users are granted organization administrator permissions only within the OCM service and this does not apply to any other Red Hat service.
- **IdP editor:** This role allows a user to manage identity providers for a cluster.
- **Machine pool editor:** This role allows a user to create, scale, and delete machine pools within a cluster.

For more detailed information about the process of adding roles to created groups, read [Managing group access with roles and members](#).

CHAPTER 6. GETTING SUPPORT FOR YOUR CLUSTERS

Learn how to contact Red Hat Support and access resources for your clusters.

6.1. OPENSIFT CONTAINER PLATFORM SUPPORT

For help with your Red Hat OpenShift Container Platform clusters, contact [Red Hat Support](#).

From here, you can:

- Open a new support case. Also see [Submitting a support case](#) in the OpenShift Container Platform documentation for instructions.
- View your open support cases: <https://access.redhat.com/support/cases/#/case/list>
- Open a live chat with support engineers
- Call or email a Red Hat Support expert

Additional resources

- See [Getting support](#) in the OpenShift Container Platform documentation for more information.

6.2. OPENSIFT DEDICATED SUPPORT

For questions about your existing Red Hat OpenShift Dedicated clusters, contact [Red Hat Support](#).

From here, you can:

- Open a new support case: <https://access.redhat.com/support/cases/#/case/>
- View open support cases: <https://access.redhat.com/support/cases/#/case/list>
- Open a live chat with support engineers
- Call or email a Red Hat Support expert

See [Support](#) in the OpenShift Dedicated documentation for more information.

6.3. RED HAT OPENSIFT SERVICE ON AWS SUPPORT

For questions about your existing Red Hat OpenShift Service on AWS (ROSA) clusters, contact [Red Hat Support](#).

From here, you can:

- Open a new support case: <https://access.redhat.com/support/cases/#/case/>
- View open support cases: <https://access.redhat.com/support/cases/#/case/list>
- Open a live chat with support engineers
- Call or email a Red Hat Support expert

See [Getting support for Red Hat OpenShift Service on AWS](#) in the ROSA documentation for more information.